Stabilizing Thompson Sampling with Null Hypothesis Bayesian Response-Adaptive Randomization

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Abstract

Response-adaptive randomization (RAR) methods can be used to adapt randomization probabilities based on accumulating data, aiming to increase the probability of allocating patients to effective treatments. A popular RAR method is Thompson sampling, which randomizes patients proportionally to the Bayesian posterior probability that each treatment is the most effective. However, its high variability can also increase the risk of assigning patients to inferior treatments and lead to inferential problems such as confidence interval undercoverage. We propose a principled method based on Bayesian hypothesis testing and model averaging to mitigate these issues: We introduce a null hypothesis postulating equal effectiveness of treatments. Bayesian model averaging then induces shrinkage toward equal randomization probabilities, with the degree of shrinkage controlled by the prior probability of the null hypothesis. Equal randomization and Thompson sampling arise as special cases when the prior probability is set to one or zero, respectively. Simulated and real-world examples illustrate that the method balances highly variable Thompson sampling with static equal randomization. A simulation study demonstrates that the method can mitigate issues with Thompson sampling and has comparable statistical properties to Thompson sampling with common ad hoc modifications such as power transformation and probability capping. We implement the method in the free and open-source R package `brar`, enabling experimenters to easily perform null hypothesis Bayesian RAR and support more effective randomization of patients.

Keywords: Adaptive trials, Bayes factor, Bayesian model averaging, outcome-adaptive randomization, spike-and-slab prior

1 Introduction

Response-adaptive randomization (RAR) methods randomly allocate patients to treatments based on accumulating data (Hu and Rosenberger, 2006; Thall and Wathen, 2007; Berry et al., 2010; Robertson et al., 2023). A popular approach is Thompson sampling (Thompson, 1933), which randomizes patients proportionally to the Bayesian posterior probability that each treatment is the most effective. Such RAR methods are attractive as they balance gathering information on treatment effectiveness and assigning subjects to effective treatments.

Despite its benefits, Thompson sampling can also introduce inferential problems such as inflated type I error rates or bias in effect estimation. Furthermore, although Thompson sampling allocates patients to the more effective treatment on average, it may also increase the probability of allocating patients to inferior treatments compared to equal randomization, particularly when treatment effects are small. This can undermine the method's intended objective and raise ethical concerns (Thall et al., 2015; Robertson et al., 2023). Consequently, there has been interest in modifying Thompson sampling to mitigate these issues.

One popular modification uses the randomization probability $\pi = p^c/\{p^c + (1-p)^c\}$, where p is the posterior probability that the experimental treatment is more effective than the control treatment, and c is a tuning parameter. Setting $c = 1$ produces Thompson sampling while $c < 1$ reduces variability. Another approach is to cap randomization probabilities, for instance, setting them to 10% or 90% if a method assigns more extreme probabilities. Finally, "burn-in" periods at the start of the study are commonly used, during which patients are randomized with equal probabilities to mitigate high variability (Wathen and Thall, 2017).

Several simulation studies have compared Thompson sampling modifications, finding that, if appropriately specified, they can mitigate its limitations (e.g., Du et al., 2017; Viele et al., 2019, 2020; Lee and Lee, 2021; Tang et al., 2025). However, such ad hoc modifications conflict with principles of coherent Bayesian learning. For example, a transformed posterior no longer corresponds to an actual posterior and cannot serve as a genuine prior for future data. This raises the question of whether a principled Bayesian RAR method can mitigate Thompson sampling's variability, and if so, how it relates to ad hoc modifications.

Principled Bayesian alternatives to Thompson sampling have also been proposed outside clinical trials, particularly in the multi-armed bandit literature. For instance, RAR based on the Gittins index prioritizes treatments with the greatest expected reward (Gittins et al., 2011), or the Bayesian upper confidence bound method allocates treatments based on posterior quantiles and offers theoretical performance guarantees (Kaufmann et al., 2012). These methods are primarily used in online decision problems but rarely in clinical trials, where additional constraints such as type I error rate control are central (Villar et al., 2015).

In this paper, we propose a novel RAR method that reduces variability in a coherent Bayesian manner, which we term "*null hypothesis Bayesian RAR*". The idea is to consider a null hypothesis postulating that treatments are equally effective. This is often equivalent to using a "spike-and-slab" prior for the treatment effect parameter, which is a mixture of a point mass at equal effectiveness and a probability density elsewhere (see e.g., Spiegelhalter et al., 2004, Section 5.5.4). The prior probability of the null hypothesis determines the mixture weight. Setting it to zero produces equal randomization, whereas setting it to one produces Thompson sampling. The method thus interpolates between equal randomization and Thompson sampling in a coherent Bayesian way. As a by-product, posterior probabilities and Bayes factors are also obtained. These can be used to monitor evidence of treatment effectiveness in a manner that is aligned with the randomization probabilities.

In the following Section 2 we introduce the general idea of the method, followed by tailoring it to normal (Section 3) and binary outcomes (Section 4). Section 5 illustrates the method on data from the ECMO trial, followed by evaluating its properties in a simulation study (Section 6). Concluding remarks are given in Section 7. The supplement illustrates our R package `brar`, and provides further details on theory and the simulation study.

2 Null hypothesis Bayesian RAR

Assume that we have observed data y and want to use these to randomize a future patient.

2.1 Two group comparisons

Suppose there is one control and one treatment group, and consider the hypotheses:

H_-: Treatment is less effective than control

H_0: Treatment and control are equally effective

H_+: Treatment is more effective than control.

How exactly these are translated into statistical hypotheses related to parameters depends on the data model, but often relates to an effect size parameter being less, equal, or greater than zero. If there is no control group, the method can still be used with the control replaced by a reference group. This hypothesis setup is similar to Cai et al. (2013) but with the addition of H_0, which introduces a different behavior of the resulting randomization probabilities.

The Bayesian posterior probability of hypothesis $H_i \in \{H_-, H_0, H_+\}$ is then

$$\Pr(H_i \mid y) = \frac{p(y \mid H_i)\Pr(H_i)}{\sum\limits_{j \in \{-,0,+\}} p(y \mid H_j)\Pr(H_j)} = 1 \Bigg/ \left\{ \sum_{j \in \{-,0,+\}} \mathrm{BF}_{ji}(y) \frac{\Pr(H_j)}{\Pr(H_i)} \right\} \tag{1}$$

where $\Pr(H_i)$ is the prior probability of H_i (with $\sum_{j \in \{-,0,+\}} \Pr(H_j) = 1$), $p(y \mid H_i) = \int_\Theta p(y \mid \theta)\, p(\theta \mid H_i)\, \mathrm{d}\theta$ is the marginal likelihood of the data y under H_i obtained from integrating the likelihood $p(y \mid \theta)$ with respect to the prior distribution $p(\theta \mid H_i)$ of the model parameters $\theta \in \Theta$ under H_i, and

$$\mathrm{BF}_{ji}(y) = \frac{\Pr(H_j \mid y)}{\Pr(H_i \mid y)} \Bigg/ \frac{\Pr(H_j)}{\Pr(H_i)} = \frac{p(y \mid H_j)}{p(y \mid H_i)} \tag{2}$$

is the Bayes factor contrasting H_j to H_i (Kass and Raftery, 1995). The Bayes factor (2) is the updating factor of the prior odds of H_j to H_i to the corresponding posterior odds (first equality), which is equivalent to the ratio of marginal likelihoods of the data under H_j and H_i (second equality). The posterior probabilities (1) can thus be computed from the marginal likelihoods of the data under each considered hypotheses along with their prior probabilities, or from the Bayes factors and prior odds relative to some reference hypothesis.

Regardless in which way they are computed, the question is how to translate posterior probabilities into randomization probabilities. We propose to randomize a future patient to the treatment group with probability

$$\pi = \Pr(H_+ \mid y) + \Pr(H_0 \mid y) \,/\, 2, \tag{3}$$

while the probability to randomize to the control group is consequently $1 - \pi = \Pr(H_- \mid y) + \Pr(H_0 \mid y) \,/\, 2$. Figure 1 shows π for different combinations of posterior probabilities. We can see that π shrinks towards 50% as the posterior probability of the null hypothesis $\Pr(H_0 \mid y)$ increases. For example, if $\Pr(H_+ \mid y) = 0.2$, $\Pr(H_- \mid y) = 0.3$, and $\Pr(H_0 \mid y) = 0.5$, as indicated by the black asterisk, the randomization probability is $\pi = 0.2 + 0.5/2 = 45\%$.

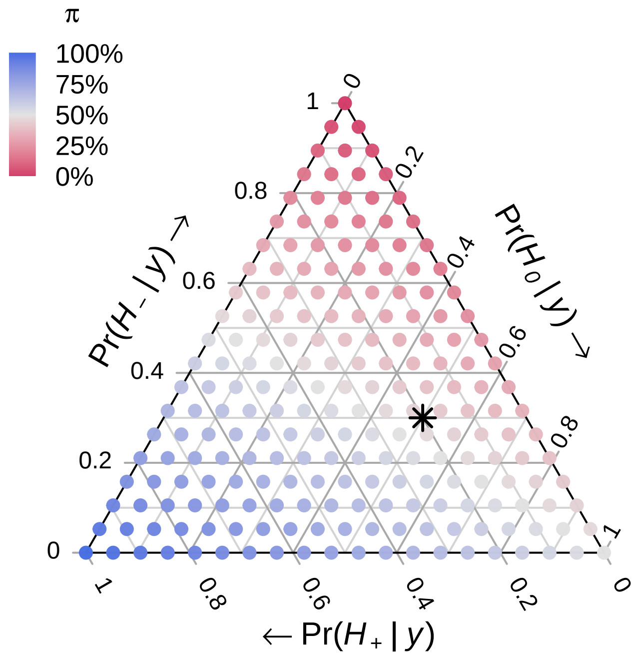

Figure 1: Ternary plot where the color of each point depicts the treatment randomization probability π for a particular combination of posterior probabilities of H_+ (bottom axis), H_- (left axis), and H_0 (right axis). For example, the black asterisk denotes the combination where $\Pr(H_+ \mid y) = 0.2$, $\Pr(H_- \mid y) = 0.3$, and $\Pr(H_0 \mid y) = 0.5$ with corresponding randomization probability $\pi = 45\%$.

This scheme can be motivated as Bayesian hypothesis averaged randomization probability where the hypothesis-specific treatment randomization probabilities are 0%, 50%, and

100%, under H_-, H_0, and H_+, respectively. That is, if we knew that H_+ or H_- is true, we should assign the next patient to the treatment or control group, respectively, to maximize patient benefit. In contrast, if we knew that H_0 is true, randomizing with $\pi = 50\%$ seems sensible.

The randomization scheme (3) reduces to Thompson sampling when the prior probability of H_0 is set to zero since then $\Pr(H_0 \mid y) = 0$ regardless of the data, and consequently $\pi = \Pr(H_+ \mid y)$. However, the scheme induces shrinkage toward $\pi = 50\%$ otherwise. In the most extreme case when $\Pr(H_0) = 1$, equal randomization is obtained as then $\Pr(H_0 \mid y) = 1$ regardless of the data, and consequently $\pi = 50\%$. The scheme thus interpolates between Thompson sampling and equal randomization in a coherent Bayesian way.

2.2 More than two groups

Suppose there are $K > 1$ treatment groups in addition to the control group. In this case, we may modify the procedure and consider the hypotheses:

H_-: All treatments are less effective than control

H_0: All treatments are equally effective as control

H_{+i}: Treatment i is more effective than control and all other treatments

for $i = 1, \dots, K$. Posterior hypothesis probabilities can still be computed from (1) but with the summation extended over all hypotheses (i.e., $j \in \{-, 0, +1, \dots, +K\}$). These can then be translated into treatment randomization probabilities

$$\pi_i = \Pr(H_{+i} \mid y) + \Pr(H_0 \mid y) \;/\; (K + 1) \tag{4}$$

and control randomization probability $1 - \sum_{i=1}^{K} \pi_i = \Pr(H_- \mid y) + \Pr(H_0 \mid y) \;/\; (K+1)$.

Also in the multi-treatment case, the randomization probabilities shrink towards equal randomization $\pi_i = 1/(K+1)$ by introducing the null hypothesis H_0. Thompson sampling and equal randomization are again obtained by setting $\Pr(H_0) = 0$ and $\Pr(H_0) = 1$, respectively, as this leads to $\Pr(H_0 \mid y) = 0$ and $\Pr(H_0 \mid y) = 1$ for any data y.

In this paper, we focus on scheme (4), which induces shrinkage toward equal randomization. However, it is important to note that there are alternatives. It may be desired to shrink to different "baseline" randomization probabilities. This can be achieved by modifying the factor of $\Pr(H_0 \mid y)$ in (4) from $1/(K+1)$ to the desired baseline randomization probability. For example, the aim may be to test all K pairwise differences between the treatments and the common control using Dunnett's test. A $\sqrt{K} : 1 : \cdots : 1$ square-root allocation of the control to treatments is commonly used to maximize the power of the test (Kieser, 2020, p.156). We may therefore use $\pi_i = \Pr(H_{+i} \mid y) + \Pr(H_0 \mid y) \,/\, (K + \sqrt{K})$. These correctly shrink RAR probabilitites towards Dunnett-type randomization probabilities, $\pi_i = 1/(K + \sqrt{K})$ for treatment $i = 1, \ldots, K$, and $1 - \sum_{i=1}^{K} \pi_i = \sqrt{K}/(K + \sqrt{K})$ for control.

2.3 Asymptotic randomization probabilities

It is of interest to understand how RAR probabilities behave as data accumulate. Under ordinary non-RAR based sampling, the posterior distribution is "consistent", meaning that as more data are generated under a hypothesis the corresponding posterior hypothesis probability tends to one (Dawid, 2011). In Section 3.1, we provide empirical support through simulation that posterior consistency also holds for the RAR setting. Thus if a treatment is superior and $\Pr(H_0) < 1$, the corresponding posterior and randomization probability tend to one. Conversely, if all treatments are equally effective and $\Pr(H_0) > 0$, the posterior probability of H_0 will approach one, and the randomization probabilities will hence approach the specified baseline randomization probabilities. This is an important difference to Thompson sampling, whose randomization probabilities do not converge to the baseline randomization probability under H_0 but remain random (Deliu and Villar, 2025).

3 Null hypothesis Bayesian RAR under normality

We now describe null hypothesis Bayesian RAR under normality.

3.1 Two group comparisons

Suppose that the data are summarized by $y = \{\hat{\theta}, \sigma\}$, where $\hat{\theta}$ is an estimate of the true treatment effect θ, and σ is the standard error of the estimate. This does not mean that the raw data, from which the estimate is computed, need to be normally distributed. For example, $\hat{\theta}$ could be an estimated (standardized) mean difference, log odds/rate/hazard ratio, risk difference, or regression coefficient. The standard error is often of the form $\sigma = \lambda/\sqrt{n}$, where n is the effective sample size and λ the standard deviation of one effective observation. Assume that the estimate is approximately normally distributed around θ with its variance σ^2 assumed to be known, i.e., $\hat{\theta} \mid \theta \sim \mathrm{N}(\theta, \sigma^2)$.

If the effect θ is oriented such that a positive effect indicates treatment benefit, the hypotheses from Section 2.1 translate into $H_-: \theta < 0$ vs. $H_0: \theta = 0$ vs. $H_+: \theta > 0$. The null hypothesis H_0 is a simple hypothesis with no free parameters, or equivalently, a point (Dirac) prior at zero. The H_- and H_+ hypotheses are composite hypotheses and require prior distributions for θ. A natural choice is a $\theta \sim \mathrm{N}(\mu, \tau^2)$ distribution whose support is truncated to the negative and positive side, respectively. In the absence of prior knowledge, it is sensible to center the prior on zero ($\mu = 0$) to represent clinical equipoise (Freedman, 1987), giving equal probability to harmful and beneficial effects. Since H_0 is a point hypothesis, the prior of θ under H_- and H_+ must be proper ($\tau < \infty$), as otherwise the posterior probability of H_0 is 1 for any data y. The value of the prior standard deviation τ must therefore be chosen so that the prior is not too vague (Kass and Raftery, 1995).

The prior probability of the null hypothesis $\Pr(H_0)$ represents the a priori plausibility of an absent effect, and also acts as a tuning parameter that controls the variability of RAR. An intuitive default is $\Pr(H_0) = 0.5$, representing equipoise between absent and present effects (Johnson, 2013). For a given $\Pr(H_0)$, it is then natural to set the prior probabilities of H_+ and H_- to $\Pr(H_+) = \{1 - \Pr(H_0)\} \times \Phi(\mu/\tau)$ and $\Pr(H_-) = \{1 - \Pr(H_0)\} \times \Phi(-\mu/\tau)$ so that to the prior distribution averaged over H_+ and H_- is again the $\mathrm{N}(\mu, \tau^2)$ distribution. For example, specifying $\Pr(H_0) = 0.5$ and a zero-centered prior, we obtain $\Pr(H_+) = \Pr(H_-) = 0.5 \times 0.5 = 0.25$. Averaging the prior over all hypotheses leads then to a spike-and-slab prior, as illustrated in Figure 2.

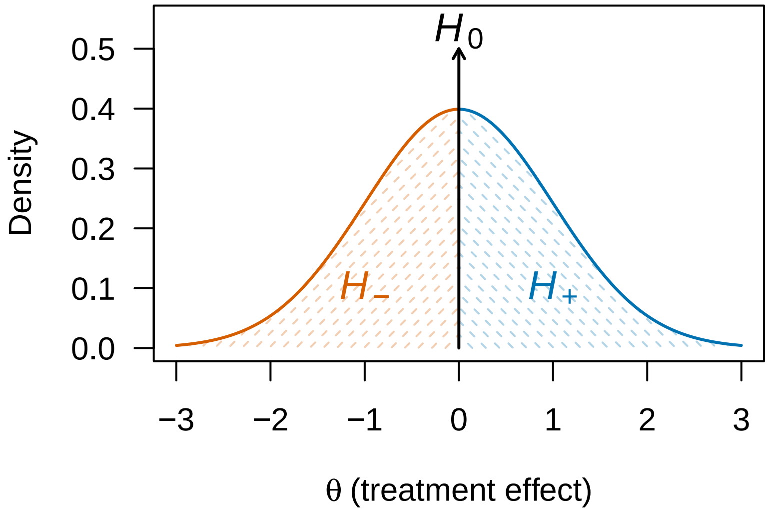

Figure 2: Illustration of spike-and-slab prior for the effect θ. A point prior at 0 is assumed under H_0. A normal prior $\theta \sim \mathrm{N}(0, 1)$ with support truncated to the positive or negative side is assumed under H_+ and H_-, respectively. These priors are averaged assuming prior hypothesis probabilities $\Pr(H_0) = 0.5$, $\Pr(H_+) = 0.25$, and $\Pr(H_-) = 0.25$.

With the priors specified, we can compute the marginal likelihood of the observed effect estimate under each hypothesis and obtain posterior probabilities (1). In the normal likelihood and prior framework, all are available in closed-form. Denoting by $\mathrm{N}(x \mid m, v)$ the normal density function with mean m and variance v evaluated at x, the marginal likelihoods are

$$\begin{aligned}
p(\hat{\theta} \mid H_-) &= \mathrm{N}(\hat{\theta} \mid \mu, \sigma^2 + \tau^2) \times \Phi(-\mu_*/\tau_*) \,/\, \Phi(-\mu/\tau) \\
p(\hat{\theta} \mid H_0) &= \mathrm{N}(\hat{\theta} \mid 0, \sigma^2) \\
p(\hat{\theta} \mid H_+) &= \mathrm{N}(\hat{\theta} \mid \mu, \sigma^2 + \tau^2) \times \Phi(\mu_*/\tau_*) \,/\, \Phi(\mu/\tau)
\end{aligned}$$

with posterior mean and variance $\mu_* = (\hat{\theta}/\sigma^2 + \mu/\tau^2)\tau_*^2$ and $\tau_*^2 = 1/(1/\sigma^2 + 1/\tau^2)$. Taking ratios of marginal likelihoods leads to the Bayes factors

$$\begin{aligned}
\mathrm{BF}_{+0}(\hat{\theta}) &= \exp\left[-\frac{1}{2}\left\{\frac{(\hat{\theta} - \mu)^2}{\sigma^2 + \tau^2} - \frac{\hat{\theta}^2}{\sigma^2}\right\}\right] \times \frac{\Phi(\mu_*/\tau_*)}{\Phi(\mu/\tau)} \Bigg/ \sqrt{1 + \frac{\tau^2}{\sigma^2}} \\
\mathrm{BF}_{+-}(\hat{\theta}) &= \frac{\Phi(\mu_*/\tau_*)}{\Phi(\mu/\tau)} \Bigg/ \frac{\Phi(-\mu_*/\tau_*)}{\Phi(-\mu/\tau)},
\end{aligned}$$

and the Bayes factors for other hypothesis comparisons can be obtained by transitivity and reciprocity (e.g., $\mathrm{BF}_{-0} = \mathrm{BF}_{+0}$ / BF_{+-}). Posterior probabilities can now be obtained by plugging the Bayes factors and prior odds into (1). Bayes factors and posterior probabilities can be monitored as data accumulate to see how the evidence for the hypotheses changes. Moreover, they can be plugged into (3) to obtain the randomization probability

$$\pi = \frac{\overbrace{\Phi\left(\mu_* / \tau_*\right)}^{\text{Thompson Sampling}}}{1 + \frac{\Pr(H_0)}{1-\Pr(H_0)} \times \mathrm{BF}_{01}(\hat{\theta})} + \frac{\overbrace{1/2}^{\text{Equal Randomization}}}{1 + \frac{1-\Pr(H_0)}{\Pr(H_0)} \,/\, \mathrm{BF}_{01}(\hat{\theta})} \tag{5}$$

where $\mathrm{BF}_{01}(\hat{\theta}) = \exp[-\{(\hat{\theta}/\sigma)^2 - (\hat{\theta} - \mu)^2/(\sigma^2 + \tau^2)\}/2]\sqrt{1 + \tau^2/\sigma^2}$ is the Bayes factor contrasting H_0 to the alternative $H_1 = H_- \cup H_+$ with prior $\theta \mid H_1 \sim \mathrm{N}(\mu, \tau^2)$. As expected, the randomization probability (5) approaches equal randomization as the prior probability of H_0 increases (i.e., $\pi \to 50\%$ as $\Pr(H_0) \nearrow 1$), whereas it approaches the ordinary Bayesian posterior tail probability of $\theta > 0$ based on a $\theta \sim \mathrm{N}(\mu, \tau^2)$ prior as the prior probability of H_0 decreases (i.e., $\pi \to 1 - \Phi\{(0 - \mu_*)/\tau_*\} = \Phi(\mu_*/\tau_*)$ as $\Pr(H_0) \searrow 0$). Similarly, the randomization probability shrinks towards 50% as the data provide more evidence for H_0 ($\pi \to 50\%$ as $\mathrm{BF}_{01} \nearrow \infty$), whereas it approaches the ordinary posterior tail probability as the evidence for H_1 increases ($\pi \to \Phi(\mu_*/\tau_*)$ as $\mathrm{BF}_{01} \searrow 0$). In contrast to ad hoc modifications of Thompson sampling, shrinkage of the posterior tail probability thus depends directly on the accumulated data via the Bayes factor.

Figure 3a shows sequences of randomization probabilities computed from normal data simulated under different true mean differences between treatment and control group θ as indicated in the panels. For each θ, one sequence was simulated under RAR using Thompson sampling ($\Pr(H_0) = 0$). RAR probabilities for other values of $\Pr(H_0)$ were computed from the same sequences to base comparison on the same data. Across all θ, the probability to randomize to the treatment group based on $\Pr(H_0) = 1$ remains static at 50% (yellow). In contrast, for a beneficial treatment effect ($\theta > 0$; left plot), the randomization probabilities based on $\Pr(H_0) < 1$ tend towards 100% as more data accumulate, whereas for a harmful treatment effect ($\theta > 0$; right plot), they tend towards 0%. Moreover, a clear ordering is

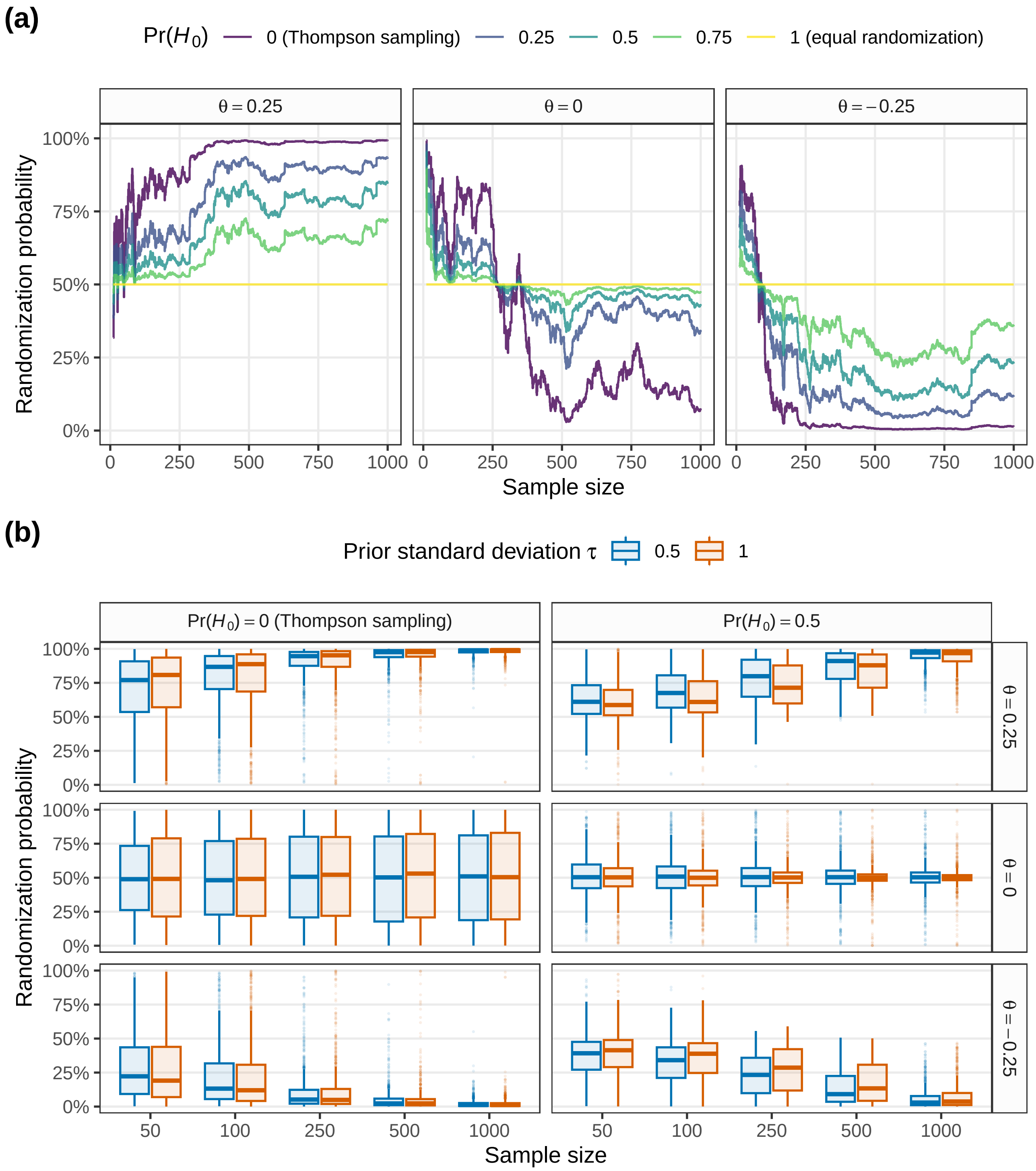

Figure 3: Evolution of Bayesian RAR probabilities for three simulated data sequences (a). In each step, one patient is allocated to the treatment or control group using Thompson sampling assuming a normal prior centered at zero with standard deviation $\tau = 1$. Patient outcomes are then simulated from a normal distribution with a true standard deviation of 1 and assuming a true mean difference θ between treatment and control group as indicated in the plot panels. Randomization probabilities under the spike-and-slab prior ($\Pr(H_0) > 0$) are computed from the same data sequence to enable comparability. Plot (b) shows the distribution of randomization probabilities based on 1′000 simulations under Bayesian RAR for two different prior standard deviations and $\Pr(H_0)$.

visible: Probabilities under $\Pr(H_0) = 0$ (Thompson sampling) are the most extreme both in the "correct" and "wrong" direction. Probabilities based on $0 < \Pr(H_0) < 1$ show the same qualitative behavior but are less extreme. Setting a higher $\Pr(H_0)$ thus reduces the variability of randomization probabilities but also slows down convergence to 100% or 0%. Finally, when there is no treatment effect ($\theta = 0$; middle plot), the probabilities based on $0 < \Pr(H_0) < 1$ stay relatively close to 50%, whereas the Thompson sampling probabilities wander more erratically.

Figure 3b shows the distribution of randomization probabilities obtained from 1′000 simulations of the scenarios from Figure 3a and for two prior standard deviations τ. When the true means differ from zero ($\theta = \pm 0.25$; first and third row), we see that the randomization probabilities for both $\Pr(H_0) = 0$ (Thompson sampling) and $\Pr(H_0) = 0.5$ correctly converge to to 100% and 0%, respectively, although the convergence of $\Pr(H_0) = 0.5$ is slower. In contrast, when there is no difference ($\theta = 0$; second row), only the randomization probabilities based on $\Pr(H_0) = 0.5$ converge to 50%, whereas the Thompson sampling randomization probabilities remain roughly uniformly distributed for all sample sizes. The qualitative behavior of the randomization probabilities is the same for both values of the prior standard deviation τ. However, convergence to 100% and 0% is slightly faster for the smaller τ (blue), whereas convergence to 50% is faster for the larger τ (orange).

3.2 More than two groups

Suppose now there are $K > 1$ treatment groups and consequently K effect estimates, each estimate quantifying the effect of the corresponding treatment relative to control. A natural generalization is to stack them into a vector $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \dots, \hat{\theta}_K)^\top$ and assume an approximate K-variate normal distribution $\hat{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \sim \mathrm{N}_K(\boldsymbol{\theta}, \boldsymbol{\Sigma})$, where $\boldsymbol{\theta} = (\theta_1, \dots, \theta_K)^\top$ is the vector of true effects and $\boldsymbol{\Sigma}$ is the (assumed to be known) covariance matrix of $\hat{\boldsymbol{\theta}}$. For example, $\hat{\boldsymbol{\theta}}$ could be a vector of estimated regression coefficients and $\boldsymbol{\Sigma}$ its estimated covariance matrix.

The hypotheses from Section 2.2 then translate into $H_-\colon\ \max\{\theta_1, \dots, \theta_K\} < 0$ vs. $H_0\colon \theta_1 = \dots = \theta_K = 0$ vs. $H_{+i}\colon \theta_i = \max\{\theta_1, \dots, \theta_K\} > 0$ for $i = 1, \dots, K$. All hypotheses apart from

H_0 are composite and require the specification of a prior distribution. In analogy to the $K = 1$ case, we specify a K-variate normal prior $\boldsymbol{\theta} \sim \mathrm{N}_K(\boldsymbol{\mu}, \boldsymbol{\mathcal{T}})$ and truncate its support to the region of the corresponding hypothesis. Similarly, a sensible default is to center the prior on $\mathbf{0}$ to encode clinical equipoise, and specify prior probabilities $\Pr(H_{+i})$ for $i = 1, \ldots, K$ so that the $\mathrm{N}_K(\boldsymbol{\mu}, \boldsymbol{\mathcal{T}})$ distribution is recovered when the prior is averaged over the hypotheses.

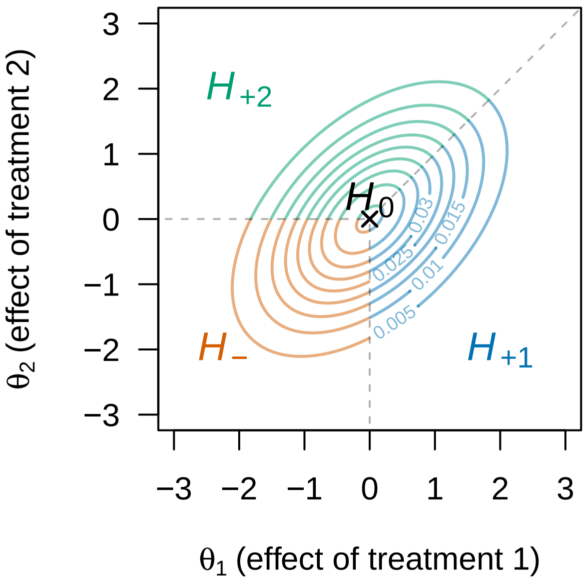

Figure 4: Illustration of a spike-and-slab prior for a two-dimensional effect $\boldsymbol{\theta} = (\theta_1, \theta_2)^\top$. A point mass prior at $(0, 0)^\top$ is assumed under H_0. A normal prior $\theta \sim \mathrm{N}((0, 0)^\top, \boldsymbol{\mathcal{T}})$ with $\boldsymbol{\mathcal{T}}_{ij} = 0.5$ for $i \neq j$ and $\boldsymbol{\mathcal{T}}_{ij} = 1$ for $i = j$, with support truncated to the space of the corresponding hypothesis is assumed under H_-, H_{+1}, and H_{+2}. The correlation of 0.5 ensures that all treatments receive equal prior probability $\Pr(H_-) = \Pr(H_{+1}) = \Pr(H_{+2}) = \{1 - \Pr(H_0)\}/3$.

Figure 4 illustrates a spike-and-slab prior for two treatment groups ($K = 2$). The prior covariance matrix $\boldsymbol{\mathcal{T}}$ is set to a uniform correlation of 0.5 which ensures that the prior probabilities of all hypotheses but H_0 are equal. Note that a zero-centered prior with uniform correlation of 0.5 ensures equal treatment prior probabilities for any number of treatment groups K, and thus is a sensible default in the absence of prior knowledege.

Also for $K > 1$, marginal likelihoods, Bayes factors, posterior probabilities, and randomization probabilities are available in closed-form (shown in the supplement). Crucially, only the evaluation of multivariate normal densities and cumulative distribution functions is required. This thus leads to an efficient Bayesian RAR method that allows experimenters to compute randomization probabilities in complex settings, for instance, multiple regression, where a full Bayesian analysis may involve various complexities, such as priors for nuisance parameters and Markov chain Monte Carlo methods for the computation of posteriors.

4 Null hypothesis Bayesian RAR for binary outcomes

Suppose now that we observe binary data of the form $y = \{y_C, y_1, \dots, y_K, n_C, n_1, \dots, n_K\}$ where y_i denotes the number of successes out of n_i trials in group $i \in \{C, 1, \dots, K\}$ coming from a control group (C) and K treatment groups. All success counts are assumed to be binomially distributed with probabilities $\theta_C, \theta_1, \dots, \theta_K$, respectively, with higher values indicating greater benefit. The hypotheses from Section 2.2 then translate into $H_-\colon \theta_C = \max\{\theta_C, \theta_1, \dots, \theta_K\}$ vs. $H_0\colon \theta_C = \theta_1 = \cdots = \theta_K$ vs. $H_{+i}\colon \theta_i = \max\{\theta_C, \theta_1, \dots, \theta_K\}$ for $i = 1, \dots, K$. In the normal framework from Section 3, these could be translated into hypotheses related to log odds ratios, which can be estimated with logistic regression or other methods. However, normal approximations can be inaccurate for small sample sizes or extreme probabilities, so exact binomial computation is preferable.

The null hypothesis H_0 requires specification of a prior for the common probability θ_C. Assuming a beta prior $\theta_C \mid H_0 \sim \text{Beta}(a_0, b_0)$, the data's marginal likelihood under H_0 is

$$\Pr(y \mid H_0) = \prod_{j \in \{C,1,\dots,K\}} \binom{n_j}{y_j} \times \frac{\mathrm{B}(a_0 + \sum_{i \in \{C,1,\dots,K\}} y_i, b_0 + \sum_{i \in \{C,1,\dots,K\}} n_i - y_i)}{\mathrm{B}(a_0, b_0)}$$

with beta function $\mathrm{B}(\cdot, \cdot)$. Under the other hypotheses, it is natural to assign independent beta priors $\theta_i \sim \text{Beta}(a_i, b_i)$ for $i \in \{C, 1, \dots, K\}$, and truncate their support to the space of the corresponding hypothesis. This leads to the marginal likelihood of the data

$$\Pr(y \mid H_i) = \prod_{j \in \{C,1,\dots,K\}} \binom{n_j}{y_j} \times \frac{\mathrm{B}(a_j + y_j, b_j + n_j - y_j)}{\mathrm{B}(a_j, b_j)} \\ \times \frac{Q_i(a_C + y_C, a_1 + y_1, \dots, a_K + y_K, b_C + n_C - y_C, b_1 + n_1 - y_1, \dots, b_K + n_K - y_K)}{Q_i(a_C, a_1, \dots, a_K, b_C, b_1, \dots, b_K)}$$

under $H_i \in \{H_C, H_{+1}, \dots, H_{+K}\}$ and with

$$Q_i(a_C, a_1, \dots, a_K, b_C, b_1, \dots, b_K) = \int_0^1 \frac{\theta_i^{a_i - 1}(1-\theta_i)^{b_i - 1}}{\mathrm{B}(a_i, b_i)} \times \prod_{j \in \{C, 1, \dots, K\} \setminus \{i\}} I_{\theta_i}(a_j, b_j) \, \mathrm{d}\theta_i,$$

where $I_x(a, b)$ is the cumulative distribution function of the beta distribution. These marginal likelihoods can be efficiently computed with numerical integration. For priors with integer hyperparameters, there even exist closed-form solutions (Howard, 1998; Miller, 2015).

For a specified $\Pr(H_0)$, we may again distribute the remaining prior probability among the other hypotheses by $\Pr(H_{+i}) = \{1 - \Pr(H_0)\} \times Q_i(a_C, a_1, \dots, a_K, b_C, b_1, \dots, b_K)$ to ensure that the averaged prior is again the beta prior. Specifying uniform priors ($a_j = b_j = 1 \; \forall \, j \in \{C, 1, \dots, K\}$) is an intuitive default that ensures that each treatment receives equal prior probability. Plugging the marginal likelihoods and prior probabilities into (1) produces posterior probabilities, which in turn can be used to obtain randomization probabilities.

5 Reanalyzing the ECMO trial

The ECMO trial (Bartlett et al., 1985; Bartlett, 2024) investigated ECMO (extracorporeal membrane oxygenation) treatment in critically ill newborns. While earlier non-randomized studies suggested a substantial treatment effect, a randomized study was required to confirm this. However, investigators were convinced the mortality risk was much higher in the control group, posing an ethical dilemma. To mitigate this, a "randomized play-the-winner" (RPW) RAR design (Wei and Durham, 1978) design was chosen.

The trial's outcome was extreme: The first newborn was randomized to ECMO and survived. The second was randomized to the control and died. All ten subsequently enrolled newborns were randomized to ECMO and survived. The trial was then stopped for efficacy, yet its unusual outcome sparked debate about whether it constituted a proper randomized trial. Later trials confirmed the efficacy of ECMO, which is now a standard treatment.

Figure 5a shows RAR probabilities computed from the ECMO data. The normal approximation and exact binomial methods were used, as well as the RPW method originally used

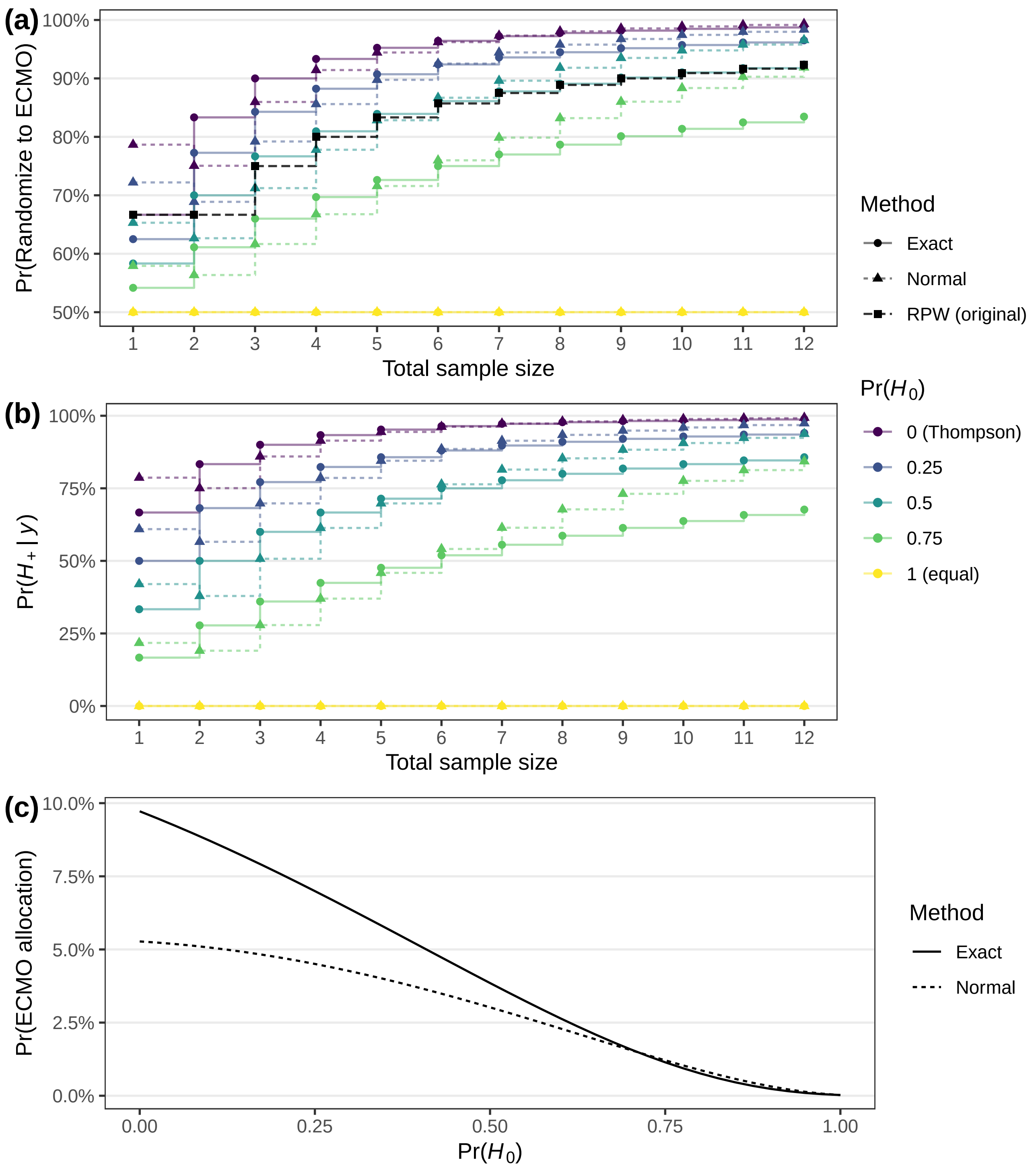

Figure 5: Evolution of Bayesian RAR randomization probabilities (a) and posterior probability of a beneficial ECMO treatment effect (b) for data from the ECMO trial. Plot (c) shows the probability of observing the ECMO allocation sequence as a function of $\Pr(H_0)$.

in the ECMO trial. A standard normal prior was assigned to the log odds ratio for the normal method, while uniform priors were assigned to the probabilities for the binomial method. Log odds ratio estimates were computed by adding a half to each cell to avoid zero cell issues. This correction induces a slight anomaly, as the probabilities from the normal method slightly decrease after the first newborn despite that it surived. This does not happen for the exact method whose probabilities increase after every additional patient. Furthermore, the randomization probabilities differ notably between the normal and exact methods, even at the end of the study. This presumably happens because the influence of the differing priors remains relatively high after observing only 12 patients.

Comparing the randomization probabilities for different prior probabilities of the null hypothesis, $\Pr(H_0) = 0$ (Thompson sampling) shows the most extreme randomization probabilities that rapidly increase to 100%, whereas $\Pr(H_0) = 1$ (equal randomization) leaves the probabilities completely static at 50%. In between, the probabilities gradually shift from 50% to the Thompson sampling probabilities. The RPW probabilities (black squares) are relatively close to the exact method with $\Pr(H_0) = 0.5$ at the beginning of the study and become closer to the normal method with $\Pr(H_0) = 0.75$ at later times.

Figure 5b shows the corresponding posterior probability of the ECMO treatment being effective. Depending on $\Pr(H_0)$, the posterior probability at the end of the trial may be very high (e.g., $\Pr(H_+ \mid y) = 0.99$ for $\Pr(H_0) = 0$), or only moderately favoring ECMO (e.g., $\Pr(H_+ \mid y) = 0.86$ for $\Pr(H_0) = 0.5$). From a Bayesian perspective, stopping the trial seems thus only a sensible decision if the prior probability of no ECMO effect was low.

Figure 5c shows the probability of observing the ECMO allocation sequence under Bayesian RAR with different $\Pr(H_0)$. The probability is highest for $\Pr(H_0) = 0$ (Thompson sampling) and decreases with increasing $\Pr(H_0)$ until $(1/2)^{12} = 0.024\%$ for $\Pr(H_0) = 1$ (equal randomization). Interestingly, the probability under RPW is also rather low (2.8%). Hence, observing the ECMO allocation sequence under Bayesian RAR would only be less likely than under RPW for $\Pr(H_0) > 0.6$ (exact method) and $\Pr(H_0) > 0.55$ (normal method), respectively.

6 Simulation study

We conducted a simulation study to evaluate the performance of null hypothesis Bayesian RAR for different values $\Pr(H_0)$, and compare it to Thompson sampling (potentially modified with burn-in periods, probability capping, power transformations), equal randomization, the Gittins index (Gittins et al., 2011), and Bayes upper confidence bound (UCB) (Kaufmann et al., 2012). Considered patient benefit measures were the mean rate of successes, the rate of extreme randomization probabilities, and sample size imbalance in favor of the inferior treatment. Bias and coverage were used to evaluate performance of response rate difference point estimates and confidence intervals under RAR, while the type I error rate and power of the corresponding tests were used to quantify hypothesis testing performance. A binomial data-generating mechanism was used. The response probability in the control group was fixed to $\theta_C = 0.25$ and varied in the first treatment group $\theta_1 \in \{0.25, 0.35, 0.45\}$ to assess the behavior of the methods for different effect sizes $\text{RD}_1 = \theta_1 - \theta_C$. In conditions with two or three treatment groups, the corresponding probabilities were set to $\theta_2 = \theta_3 = 0.3$. More detailed descriptions of the design and results of the simulation study are provided in the supplement and an online results dashboard (`https://samch93.github.io/brar/`).

A trade-off between patient benefit and inferential operating characteristics was observed across most settings. Methods that improved patient benefit through more aggressive allocation (Gittins index, Bayes UCB, and Thompson sampling) often had worse bias, coverage, type I error rate, and power, whereas more conservative methods tended to preserve inferential validity at the cost of reduced patient benefit. Figure 6 illustrates this trade-off for the mean success rate and coverage. This trade-off is well documented (Hu and Rosenberger, 2003; Zhang and Rosenberger, 2005; Williamson and Villar, 2019; Robertson et al., 2023). However, there were also settings with a small "window" in which patient benefit could vary while inferential performance remained essentially unchanged. For example, with a burn-in of 50, two treatment groups ($K = 2$), and a large effect size ($\text{RD}_1 = 0.2$), coverage was roughly constant across $\Pr(H_0)$, whereas the mean success rate changed noticeably. Increasing $\Pr(H_0)$ reduced imbalance toward the inferior treatment, though negative imbalance was also strongly influenced by effect size. In conditions with large effect size ($\mathrm{RD}_1 = 0.2$), the proportion of negative imbalance was often lower for RAR methods (including Thompson sampling) than equal randomization, thus replicating the results from Robertson et al. (2023). These findings highlight the importance of evaluating operating characteristics during trial design, since appropriately specified RAR designs may enable improved patient benefit without substantial loss of inferential performance.

Under most conditions, Bayesian RAR with $\Pr(H_0) = 0.75$ showed similar operating characteristics to Thompson sampling with capped randomization probabilities at 10%/90% and power transformation $c = i/(2n)$, where i is the current sample size and n is the maximum sample size. Both mitigated issues with ordinary Thompson sampling, exhibiting less negative sample size imbalance, reduced bias, improved coverage, and lowered type I error rates. This came at the cost of worse patient benefit performance, for instance, lower mean success rate, though still better than equal randomization in most cases. Lower values of $\Pr(H_0)$ produced operating characteristics comparable to less extreme modifications, such as $c = 1/2$. Finally, the results were similar for exact binomial and approximate normal versions of the method (supplement), suggesting some robustness to prior and model choice.

While these results demonstrated that null hypothesis Bayesian RAR has comparable properties to capped Thompson sampling for realistic trial sizes, it is important to note that the two are expected to differ asymptotically. Capped Thompson sampling enforces bounds on randomization probabilities, whereas uncapped Bayesian RAR probabilities can converge to 100% or 0%. The observed similarity is thus only a finite-sample phenomenon.

7 Discussion

We have proposed a modification of Thompson sampling by recasting the problem in a Bayesian hypothesis testing framework and introducing a null hypothesis of equal effectiveness. This method can interpolate between equal randomization and Thompson sampling by changing the prior probability of the null hypothesis $\Pr(H_0)$. This allows balancing

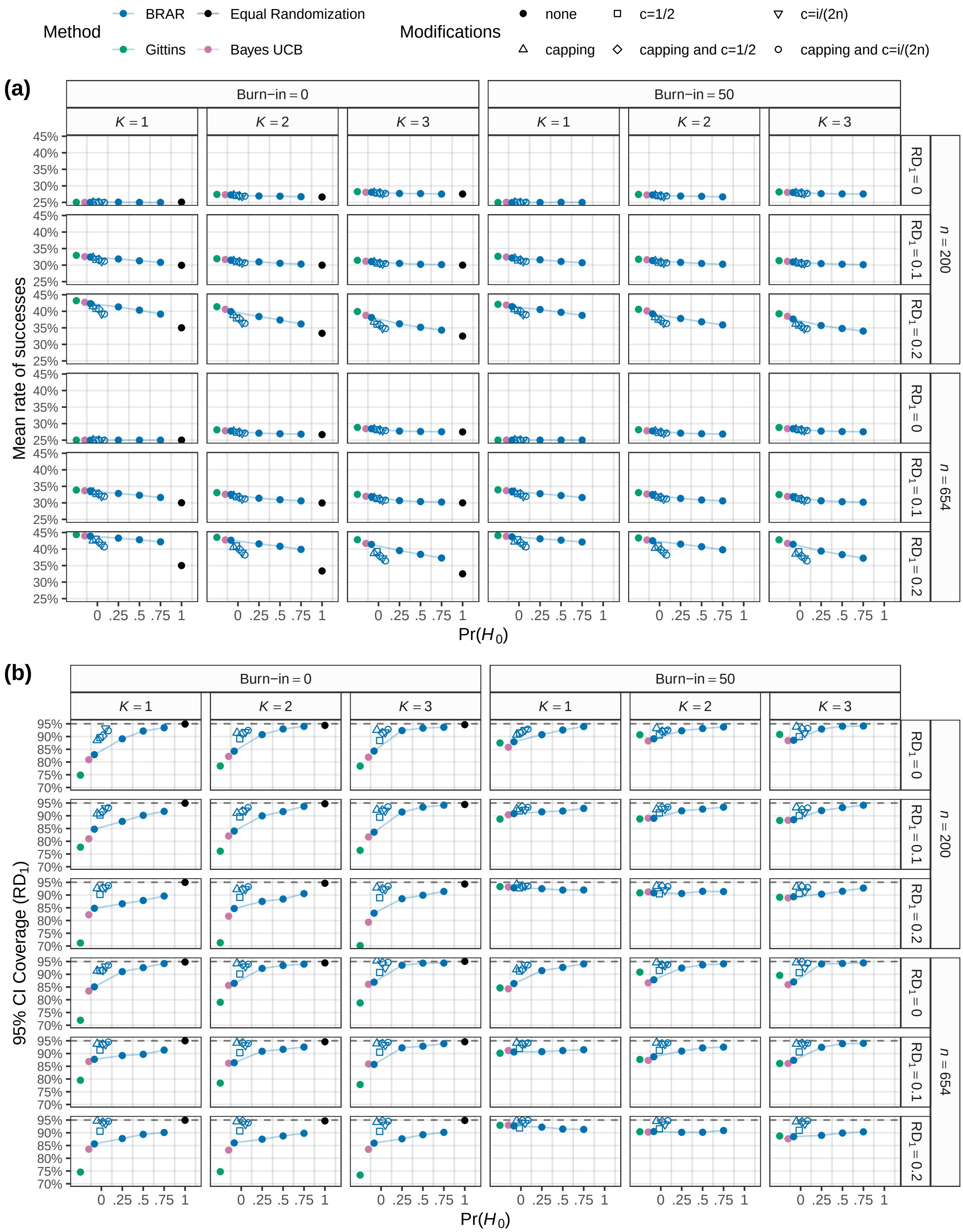

Figure 6: Plot (a) shows the mean rate of successes (i.e., the number of successes in a study divided by its sample size averaged over all 10’000 simulation repetitions). The maximum Monte Carlo standard error (MCSE) is 0.051%. Plot (b) shows the empirical coverage rate of the 95% Wald confidence interval for the response rate difference RD_1. The maximum MCSE is 0.46%.

patient benefit with inferential properties. For large values of $\Pr(H_0)$, we observed similar characteristics as with ad hoc modifications of Thompson sampling.

One advantage of the method is that randomization probabilities coherently correspond with the available statistical evidence (in the form of Bayes factors) and beliefs (in the form of posterior probabilities). Both could also serve as decision-making tools instead of frequentist criteria. For example, a study could be stopped if the posterior probability of efficacy exceeds 0.99. This also aligns with the perspective that it is unnatural to randomize patients using extreme randomization probabilities associated with such high posterior probabilities.

Although we conducted a simulation study to understand the method's basic behavior, more realistic evaluations are needed to understand its real-world applicability. For example, the method needs to be evaluated in combination with futility stopping (e.g., dropping of treatment arms which are shown to be ineffective at interim) and under time trends (e.g., treatment effects attenuating over time because the standard of care improves). Regarding prior choice, $\Pr(H_0) = 0.75$ mitigated many Thompson sampling issues in our study, yet higher values may also be considered. Similar considerations apply to the prior distributions of the parameters, which we only limitedly varied. For instance, rather than setting the correlation of the multivariate normal prior to achieve equal randomization probabilities, it could be beneficial to set it so that the prior probability of the control being superior is always $\Pr(H_0) = 0.5$ and the remaining probability is distributed equally among the treatments. Similarly, concurrent allocation to the control group, as in Trippa et al. (2012), may be desirable. Future work may therefore investigate whether more efficient RAR can be obtained by tuning the prior distribution. RAR is often impractical when outcomes require long follow-up. An alternative could be to perform Bayesian RAR with a surrogate outcome that is sooner observed than the primary outcome (Gao et al., 2024). Finally, further theoretical work is needed to understand the asymptotic behavior of the method. Most importantly, the empirically observed convergence of randomization probabilities to 50% under H_0 remains to be shown rigorously.

Acknowledgments and conflict of interest We thank František Bartoš and Małgorzata Roos for valuable comments. We thank the associate editor and two anonymous reviewers for excellent suggestions. We declare no conflict of interest.

Data availability Code and data to reproduce our results are available at `https://github.com/SamCH93/brar` and archived at `https://doi.org/10.5281/zenodo.17248628`.

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Appendix A The R package brar

Our R package can be installed by running `remotes::install_github(repo = "SamCH93/brar", subdir = "package")` in an R session (requires the `remotes` package, which is available on CRAN). The main functions of the package are `brar_normal` and `brar_binomial`, which implement the approximate normal method from Section 3 and the exact binomial method from Section 4 in the main text. The following code chunk illustrates how the latter function can be used.

```
library(brar) # load package

## observed successes and trials in control and 3 treatment groups
y <- c(10, 9, 14, 13)
n <- c(20, 20, 22, 21)

## conduct exact point null Bayesian RAR
brar_binomial(y = y, n = n,
              ## uniform prior for common probability under H0
              a0 = 1, b0 = 1,
              ## uniform priors for all probabilities
              a = c(1, 1, 1, 1), b = c(1, 1, 1, 1),
              ## prior probability of the null hypothesis
              pH0 = 0.5)

## DATA
##             Events Trials Proportion
## Control        10     20      0.500
## Treatment 1     9     20      0.450
## Treatment 2    14     22      0.636
## Treatment 3    13     21      0.619
##
## PRIOR PROBABILITIES
##    H-     H0    H+1    H+2    H+3
## 0.125 0.500 0.125 0.125 0.125
##
## BAYES FACTORS (BF_ij)
##          H-      H0    H+1     H+2    H+3
```

```
## H-   1.000 0.0341  2.16 0.1837 0.223
## H0  29.335 1.0000 63.45 5.3891 6.533
## H+1  0.462 0.0158  1.00 0.0849 0.103
## H+2  5.443 0.1856 11.77 1.0000 1.212
## H+3  4.490 0.1531  9.71 0.8249 1.000
##
## POSTERIOR PROBABILITIES
##      H-      H0     H+1     H+2     H+3
## 0.00777 0.91148 0.00359 0.04228 0.03488
##
## RANDOMIZATION PROBABILITIES
##     Control Treatment 1 Treatment 2 Treatment 3
##       0.236       0.231       0.270       0.263
```

Appendix B Null hypothesis Bayesian RAR under multivariate normality

Suppose there are $K > 1$ treatment groups and consequently K effect estimates, each estimate quantifying the effect of the corresponding treatment relative to the control. A natural generalization univariate normal Bayesian RAR is to stack the estimates into a vector $\hat{\boldsymbol{\theta}} = (\hat{\theta}_1, \ldots, \hat{\theta}_K)^\top$ and assume an approximate K-variate normal distribution $\hat{\boldsymbol{\theta}} \mid \boldsymbol{\theta} \sim \mathrm{N}_K(\boldsymbol{\theta}, \boldsymbol{\Sigma})$, where $\boldsymbol{\theta} = (\theta_1, \ldots, \theta_K)^\top$ is the vector of true effects and $\boldsymbol{\Sigma}$ is the (assumed to be known) covariance matrix of $\hat{\boldsymbol{\theta}}$. For example, $\hat{\boldsymbol{\theta}}$ could be a vector of estimated regression coefficients and $\boldsymbol{\Sigma}$ the corresponding covariance matrix.

The hypotheses from Section 2.2 in the maini text then translate into

$$H_-\colon \max\{\theta_1, \ldots, \theta_K\} < 0 \quad \text{vs.} \quad H_0\colon \theta_1 = \cdots = \theta_K = 0 \quad \text{vs.} \quad H_{+i}\colon \theta_i = \max\{\theta_1, \ldots, \theta_K\} > 0$$

for $i = 1, \ldots, K$. All hypotheses apart from H_0 are composite and require the specification of a prior distribution. In analogy to the one treatment case, we specify a K-variate normal prior $\boldsymbol{\theta} \sim \mathrm{N}_K(\boldsymbol{\mu}, \boldsymbol{\mathcal{T}})$ and truncate its support to the region of the corresponding hypothesis (e.g., for hypothesis H_{+i}, the space in $\mathbb{R}^K$ where the ith component is positive and larger than and all other components). As in the one group case, it seems a sensible default to center the prior on $\mathbf{0}$ to encode clinical equipoise. Similarly, the prior hypothesis probabilities $\Pr(H_{+i})$ for $i = 1, \ldots, K$ may again be specified so that the $\mathrm{N}_K(\boldsymbol{\mu}, \boldsymbol{\mathcal{T}})$ distribution is recovered when the prior is averaged over the hypotheses. Additionally, specifying a prior covariance matrix $\boldsymbol{\mathcal{T}}$ with uniform correlation of 0.5 ensures equal treatment prior probabilities for any number of treatment groups K, and thus seems a sensible default in the absence of prior knowledege.

The normal-normal conjugate framework allows us to derive marginal likelihoods in closed-form. The marginal likelihood under H_0 is

$$p(\hat{\boldsymbol{\theta}} \mid H_0) = \mathrm{N}_K(\hat{\boldsymbol{\theta}} \mid \mathbf{0}, \boldsymbol{\Sigma}),$$

while the marginal likelihood under H_- is

$$p(\hat{\boldsymbol{\theta}} \mid H_-) = \mathrm{N}_K(\hat{\boldsymbol{\theta}} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma} + \boldsymbol{\mathcal{T}}) \times \frac{\Phi_K(\mathbf{0} \mid \boldsymbol{\mu}_*, \boldsymbol{\mathcal{T}}_*)}{\Phi_K(\mathbf{0} \mid \boldsymbol{\mu}, \boldsymbol{\mathcal{T}})}$$

with $\mathrm{N}_K(\boldsymbol{x} \mid \boldsymbol{m}, \boldsymbol{V})$ and $\Phi_K(\boldsymbol{x} \mid \boldsymbol{m}, \boldsymbol{V})$ the density and cumulative distribution functions of the K-variate normal distribution with mean vector $\boldsymbol{m}$ and covariance matrix $\boldsymbol{V}$ evaluated at $\boldsymbol{x}$, and posterior mean $\boldsymbol{\mu}_* = (\boldsymbol{\Sigma}^{-1} + \boldsymbol{\mathcal{T}}^{-1})^{-1}(\boldsymbol{\Sigma}^{-1}\hat{\boldsymbol{\theta}} + \boldsymbol{\mathcal{T}}^{-1}\boldsymbol{\mu})$ and covariance $\boldsymbol{\mathcal{T}}_* = (\boldsymbol{\Sigma}^{-1} + \boldsymbol{\mathcal{T}}^{-1})^{-1}$. Finally, the marginal likelihood under H_{+i} is given by

$$p(\hat{\boldsymbol{\theta}} \mid H_{+i}) = \mathrm{N}_K(\hat{\boldsymbol{\theta}} \mid \boldsymbol{\mu}, \boldsymbol{\Sigma} + \boldsymbol{\mathcal{T}}) \times \frac{\Phi_K(\mathbf{0} \mid \boldsymbol{A}_{i,K}\boldsymbol{\mu}_*, \boldsymbol{A}_{i,K}\boldsymbol{\mathcal{T}}_*\boldsymbol{A}_{i,K}^\top)}{\Phi_K(\mathbf{0} \mid \boldsymbol{A}_{i,K}\boldsymbol{\mu}, \boldsymbol{A}_{i,K}\boldsymbol{\mathcal{T}}\boldsymbol{A}_{i,K}^\top)}$$

where $\boldsymbol{A}_{i,K}$ is a $K \times K$ contrast matrix that maps $\boldsymbol{\theta}$ to the space where the negative orthant corresponds to the space of hypothesis H_{+i}. For example, for $i = 2$ and $K = 3$, the matrix is

$$\boldsymbol{A}_{2,3} = \begin{bmatrix} 0 & -1 & 0 \\ 1 & -1 & 0 \\ 0 & -1 & 1 \end{bmatrix}$$

with the first row encoding the constraint of θ_2 being positive, and the second and third rows encoding the constraints of θ_2 being larger than θ_1 and θ_3, respectively. As expected, for $K = 1$, these marginal likelihoods reduce to the univariate ones (7) in the main text. Moreover, since they are all available in closed-form, Bayes factors, posterior probabilities, and randomization probabilities are also available in closed-form. Crucially, only the exact evaluation of multivariate normal densities and cumulative distribution functions is required, and both are efficiently implemented in statistical software (e.g., in the `mvtnorm` R package, Genz and Bretz, 2009). This thus leads to an efficient Bayesian RAR method that allows experimenters to compute randomization probabilities in complex settings, for instance, multiple regression, where a full Bayesian analysis may involve various complexities, such as priors for nuisance parameters and Markov chain Monte Carlo methods for the computation of posteriors.

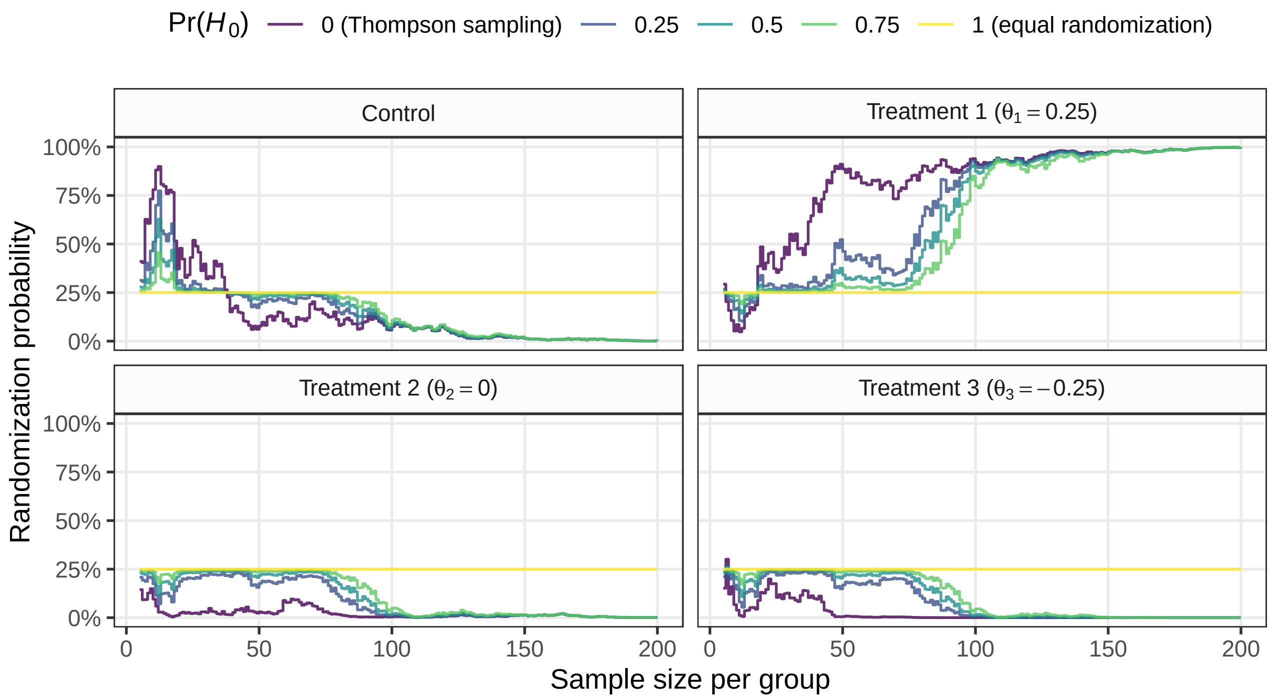

Figure S1: Evolution of Bayesian RAR probabilities for 3 treatments and a control group. In each step, one observation from each group is simulated from a normal distribution with a standard deviation of 1 and corresponding mean differences as indicated in the panel titles (treatment 1 is the most effective treatment), and a mean difference effect estimate $\hat{\boldsymbol{\theta}}$ and its estimated covariance matrix $\boldsymbol{\Sigma}$ computed with linear regression. Randomization probabilities are then computed using a normal spike-and-slab prior for centered at the origin and with covariance matrix $\mathcal{T}$ with $\mathcal{T}_{ij} = 0.5$ for $i \neq j$ and $\mathcal{T}_{ij} = 1$ for $i = j$, and for different prior probabilities $\Pr(H_0)$.

Figure S1 shows sequences of randomization probabilities computed from simulated normal data with $K = 3$ treatment groups. We see that assigning a prior probability $\Pr(H_0) =$

1 leads to static equal randomization at $\pi_i = 1/(K+1) = 25\%$, while assigning $\Pr(H_0) = 0$ (Thompson sampling) leads to the most variable randomization probabilities. Since data are simulated assuming that treatment 1 is the most effective, randomization probabilities based on $\Pr(H_0) < 1$ converge towards $\pi_1 = 100\%$ and toward 0% for the remaining treatments. While convergence is the fastest for $\Pr(H_0) = 0$, this prior probability also accidentally produces rather high randomization probabilities for the control group at the start of the study, which is less pronounced for positive prior probabilities $\Pr(H_0) > 0$.

Appendix C Simulation study

We now describe the design and results of our simulation study following the structured ADEMP approach (Morris et al., 2019; Siepe et al., 2024). Our simulation study was not preregistered as it constitutes early-phase methodological research where the properties of a new method are explored without the intention to give wide recommendations for practitioners (Heinze et al., 2023). A website with additional details and results is provided at `https://samch93.github.io/brar/`.

C.1 Aims

The aim of the simulation study is to evaluate the design characteristics of the newly proposed null hypothesis Bayesian RAR approach, and compare it to existing methods.

C.2 Data-generating mechanism

The data-generating mechanism was inspired by the simulation studies from Robertson et al. (2023), Thall and Wathen (2007), and Wathen and Thall (2017). In each repetition, a data set with n binary outcomes is simulated through RAR: A patient i is randomly allocated to the control group or one of the K treatment groups based on randomization probabilities computed from the $1, \dots, i-1$ preceding outcomes. Depending on the allocation, an outcome is either simulated from a Bernoulli distribution with probability θ_C in the control group, θ_1 in the first treatment group, or θ_2 for the remaining treatment groups (in case $K > 1$).

Parameters were chosen similar to the simulation study from Robertson et al. (2023). We vary the sample size $n \in \{200, 654\}$ to represent low and high powered studies, the number of treatment groups $K \in \{1, 2, 3\}$, and the probability in the first treatment group $\theta_1 \in \{0.25, 0.35, 0.45\}$. The probability in the control group and the remaining groups is always fixed at $\theta_C = 0.25$ and $\theta_2 = \theta_3 = 0.3$, respectively. All these parameters are varied fully-factorially, leading to $2 \times 3 \times 3 = 18$ parameter conditions.

Since treatment allocation determines from which true probability an outcome is simulated, data generation is directly influenced by the RAR methods described below. These come with additional parameters that are, however, considered as method tuning parameters rather than true underlying parameters.

C.3 Estimands and other targets

The primary interest of this simulation study lies in assessing the patient benefit characteristics of different RAR methods. Additionally, the estimand of interest is the response rate difference $\text{RD}_1 = \theta_1 - \theta_C$ and the target of interest is the null hypothesis of $\text{RD}_1 = 0$.

C.4 Methods

We consider the null hypothesis Bayesian RAR method described in the main text. The prior probability of H_0 is a tuning parameter and controls the variability of the randomization probabilities. Setting $\Pr(H_0) = 1$ produces equal randomization, whereas $\Pr(H_0) = 0$ produces Thompson sampling. We consider values of $\Pr(H_0) \in \{0, 0.25, 0.5, 0.75, 1\}$, as well as the normal approximation and exact binomial version of RAR. For approximate normal RAR, a normal prior with mean 0, variance 1, and in case of $K > 1$ a correlation of 0.5, is considered. Independent uniform priors are assigned for binomial RAR. Log odds ratios along with their covariance are estimated with logistic regression and then used as inputs for the normal RAR method, while the exact method uses success counts and sample sizes only. In case a method fails to converge, equal randomization is applied as a back-up strategy, as this mimics what an experimenter might do in practice when a RAR method fails to converge (Pawel et al., 2025).

We also consider three modifications of these methods: In some conditions, a "burn-in" phase is carried out during which the first 50 patients are always randomized with equal probability $1/(K+1)$ to each group. For Thompson sampling ($\Pr(H_0) = 0$), we additionally consider conditions with power transformations of randomization probabilities, i.e., if π_k is the randomization probability of group k, we take $\pi_k^* = \pi_k^c / \sum_{j \in \{C,1,\dots,K\}} \pi_j^c$. We consider $c = 1/2$ and $c = i/(2n)$ with i the current and n the maximal sample size, which are two popular choices of the tuning parameter c (Wathen and Thall, 2017). Additionally, in some conditions "capping" is applied to Thompson sampling. That is, randomization probabilities outside the $[10\%, 90\%]$ interval are set to either 10% or 90%. After capping has been performed, randomization probabilities are re-normalized to sum to one (Wathen and Thall, 2017; Lee and Lee, 2021). This re-normalization is only performed for randomization probabilities greater than 10%, as these would otherwise be reduced again to probabilities less than 10%. In case, a re-normalized probability becomes less than 10%, it is also capped at 10% and a second re-normalization performed. For equal randomization ($\Pr(H_0) = 1$), no burn-in, capping, or power transformation conditions are simulated as these manipulations have no effect.

Finally, upon suggestions from the associate editor, we also include two methods from the multi-armed bandit literature in the simulation study to compare null hypothesis Bayesian RAR to more aggressive allocation methods – the Gittins index (Gittins et al., 2011; Villar et al., 2015) and the Bayesian upper confidence bound (Bayes UCB) methods (Kaufmann et al., 2012). For the Gittins index, we use the implementation from the `RARtrials` R package (Xu et al., 2025) with a discount factor of $d = 0.995$. For Bayes UCB we use uniform priors for each probability along with the tuning parameter $c = 0$, as recommended in Kaufmann et al. (2012) based on their simulation results.

C.5 Performance measures

Patient benefit was quantified with:

- The mean rate of successes per study

$$\overline{\text{RS}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \sum_{j=1}^{n} \frac{y_{ij}}{n}$$

  where y_{ij} denotes the $0/1$ success indicator of patient j in simulation i, n is the sample size, and n_{sim} is the number of simulation repetitions.

- The mean rate of extreme randomization probabilities (less than 10% or greater than 90%)

$$\overline{\text{REP}} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \sum_{j=1}^{n} \frac{\mathbb{1}(\text{any randomization probability at time } j \; < 10\% \text{ or } > 90\%)}{n}$$

  with indicator function $\mathbb{1}(\cdot)$.

- The proportion of simulations where the number of allocations to treatment 1 was at least 10% of the total sample size n less than the average sample size in the remaining groups

$$\hat{S}_{0.1} = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \mathbb{1}\left(\frac{n - n_{1i}}{K} - n_{1i} > 0.1n\right)$$

  where n_{1i} is the number of allocations to treatment group 1 in simulation i. For $K = 1$, this reduces to the $\hat{S}_{0.1}$ sample size imbalance measure from Robertson et al. (2023), which in turn was inspired by the performance evaluation in the simulation study from Thall et al. (2015). Note that the $\hat{S}_{0.1}$ measure is difficult to translate into an actual patient benefit measure because the patient benefit loss related to imbalance depends on the size of the treatment effect (Robertson et al., 2023). For instance, imbalance is less consequential if the treatment effect is small than if it is large. Nevertheless, we include the $\hat{S}_{0.1}$ measure for comparability with previous simulation studies.

Parameter estimation and hypothesis testing performance was quantified with:

- The empirical bias of the estimate of the rate difference RD_1

$$\mathrm{Bias}(\mathrm{RD}_1) = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \hat{\theta}_{1i} - \hat{\theta}_{0i} - \mathrm{RD}_1$$

  where $\hat{\theta}_{1i}$ and $\hat{\theta}_{Ci}$ are the maximum likelihood estimates of the probabilities θ_1 and θ_C in simulation repetition i.

- Empirical coverage of the 95% Wald confidence intervals of RD_1

$$\mathrm{Coverage}(\mathrm{RD}_1) = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \mathbb{1}\left\{95\% \text{ CI includes } \mathrm{RD}_1 \text{ in simulation } i\right\}.$$

- Empirical rejection rate (type I error rate or power depending on the condition) related to the Wald test of the rate difference RD_1

$$\mathrm{RR}(\mathrm{RD}_1) = \frac{1}{n_{\text{sim}}} \sum_{i=1}^{n_{\text{sim}}} \mathbb{1}\left\{\text{Test rejects } H_0\colon \mathrm{RD}_1 = 0 \text{ in simulation } i\right\}.$$

Each condition was simulated 10'000 times. This ensures a MCSE (Monte Carlo Standard Error) for type I error rate, power, and coverage of at most 0.5%. MCSEs were calculated using the formulae from Siepe et al. (2024) and are provided for all measures in the following figures and the supplemental website.

C.6 Computational aspects

The simulation study was run on a server running Debian GNU/Linux forky/sid and R version 4.5.2 (2025-10-31). The SimDesign R package was used to organize and run the simulation study (Chalmers and Adkins, 2020). Our newly developed `brar` R package was used to perform Bayesian RAR. Code and data to reproduce this simulation study are available at `https://github.com/SamCH93/brar` and persistently archived at `https://doi.org/10.5281/zenodo.17248628`.

C.7 Results

Convergence Non-convergence happened only rarely for the normal approximation method due to logistic regression not converging at the start of the study when no events were observed for some groups. In this case, equal randomization was applied. The highest rate of such non-convergence was in a condition with $K = 3$ treatment groups where 4.6% of the n logistic regressions did not converge. No other forms of missingness were observed. Figures and tables with per-condition-method non-convergence rates are available at https://samch93.github.io/brar/.

Rate of successes and extreme randomization probabilities The mean rate of successes is shown in Figure S2. It was generally the highest for the Gittins index, followed by Bayes UCB, Thompson sampling, and being the lowest for equal randomization. The normal approximation and exact method produced mostly similar rates, with the normal method sometimes showing slightly higher rates (e.g., for $K = 3$ and $\text{RD}_1 = 0.2$). The Thompson sampling modifications generally reduced the mean success rate, with the greatest reduction achieved by combining capping and power transformation with $c = i/(2n)$. In conditions with small sample size these rates were similar as when the prior probability was $H_0 = 0.75$, while they were lower when the sample size was larger. This makes sense as for larger sample sizes, uncapped randomization probabilities are more likely to converge to extreme ones. This is also visible in Figure S3 where more extreme randomization probabilities are observed with increasing sample size and rate difference for all methods but the ones with capping. Finally, Bayes UCB and the Gittins index methods always produced extreme randomization probabilities, as these methods deterministically allocate the next patient to the group with the highest upper confidence bound for and Gittins index, respectively.

Negative sample size imbalance Figure S4 shows negative sample size imbalance as quantified by the $\hat{S}_{0.1}$ metric. The Gittins index generally showed the highest negative imbalance, followed by Thompson sampling and Bayes UCB. Among the methods with $\Pr(H_0) < 1$, negative imbalance was the greatest for Thompson sampling and reduced when

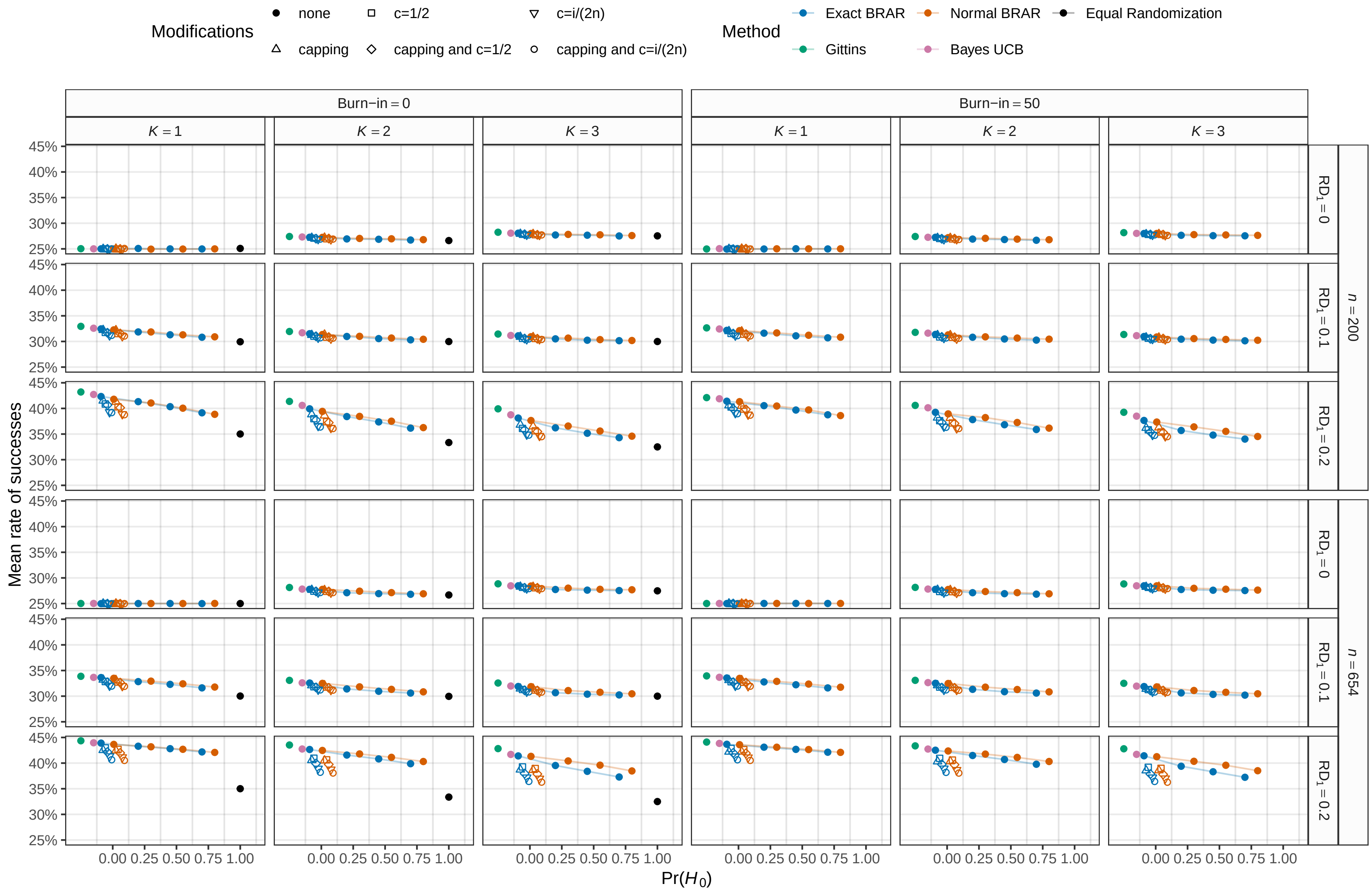

Figure S2: Mean rate of successes (i.e., the number of successes in a study divided by its sample size averaged over all 10′000 simulation repetitions). The maximum MCSE is 0.051%.

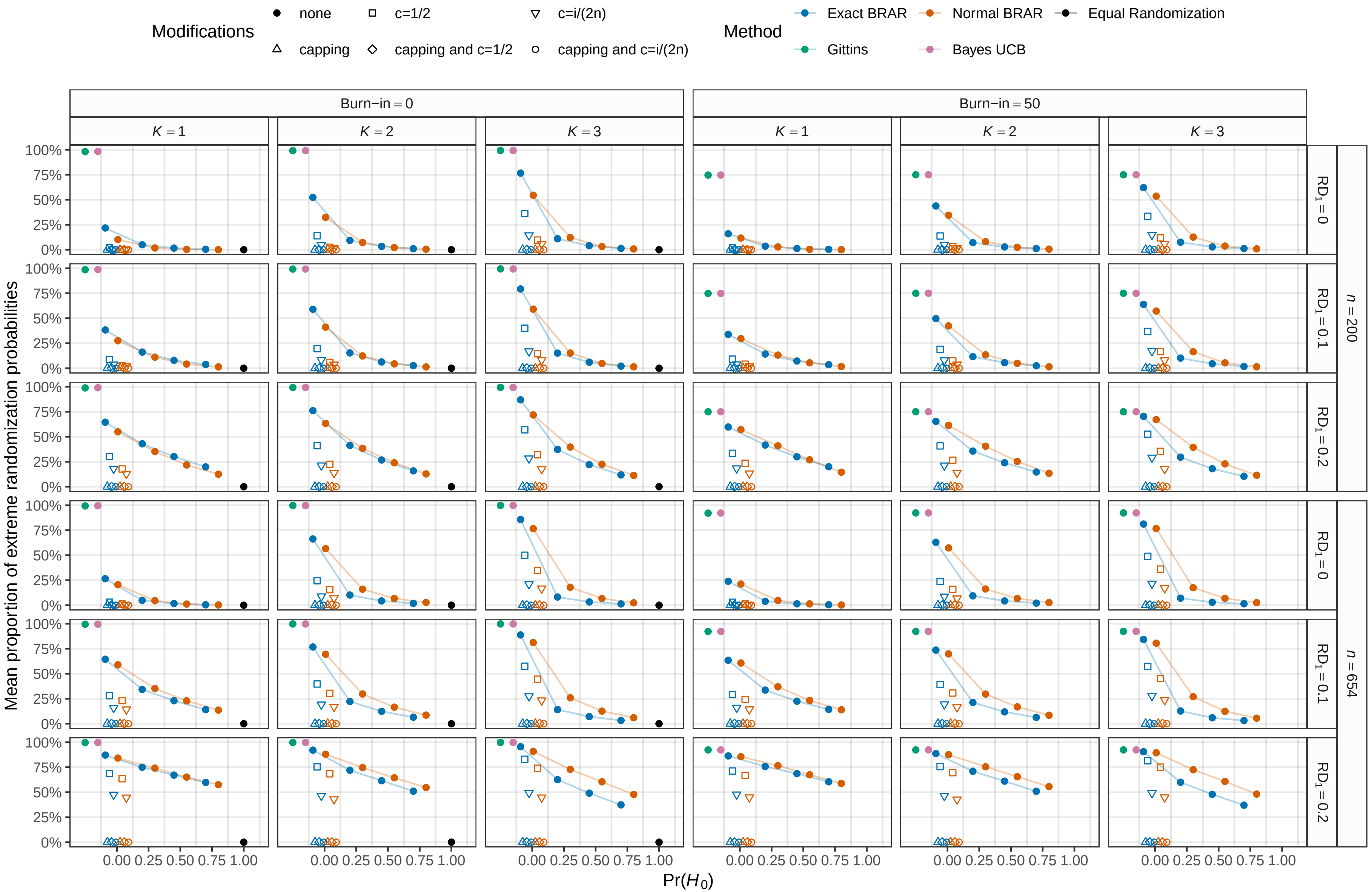

Figure S3: Mean proportion of 10′000 simulations with randomization probabilities either less than 10% or greater than 90%. The maximum MCSE is 0.33%.

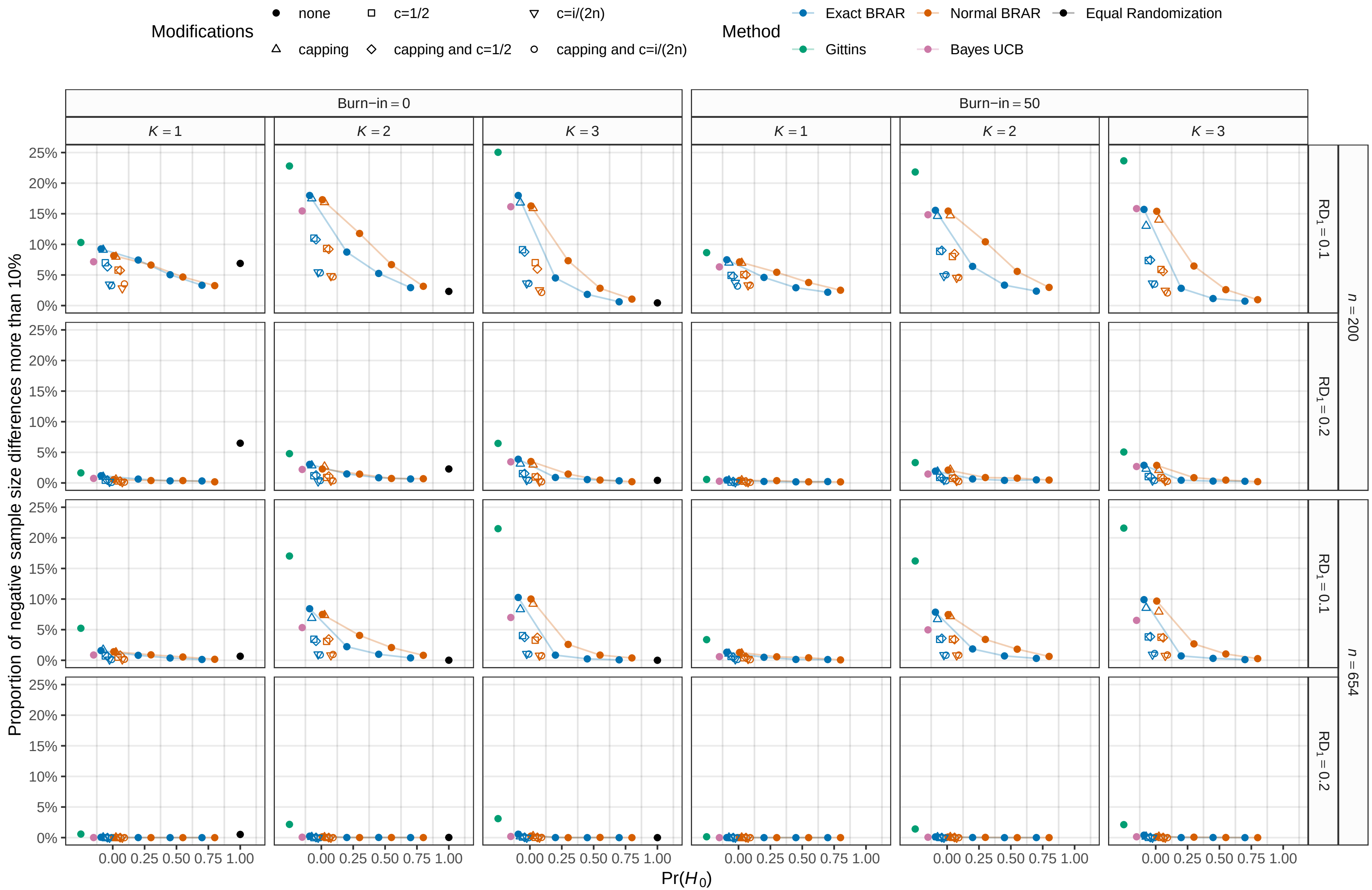

Figure S4: Proportion of 10′000 simulations with more than 10% of the sample size randomized to other groups than treatment group 1. The maximum MCSE is 0.5%.

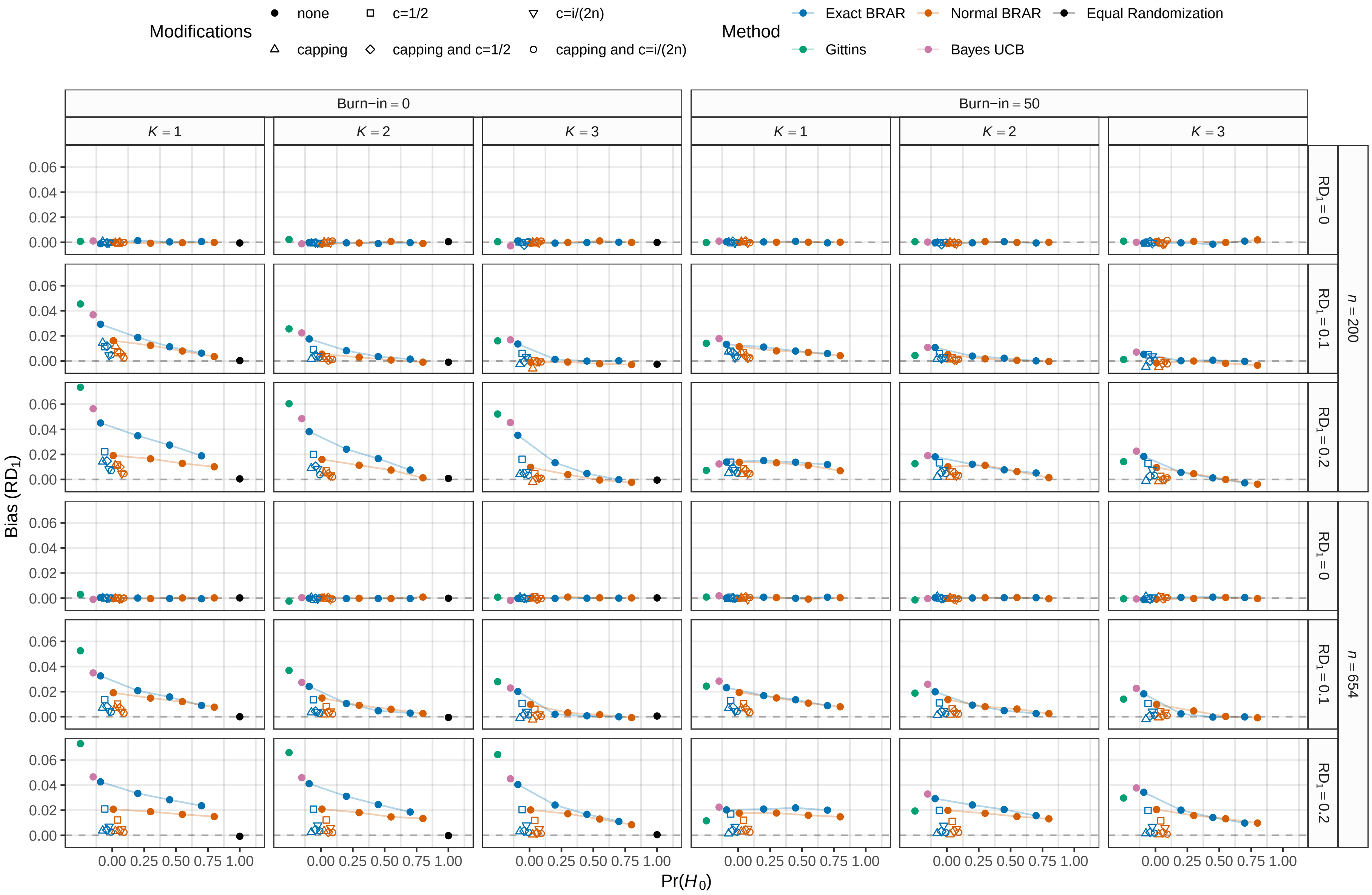

Figure S5: Empirical bias of the estimate of the rate difference RD_1 between the first treatment group and the control group based on 10′000 simulation repetitions. The maximum MCSE is 0.0016.

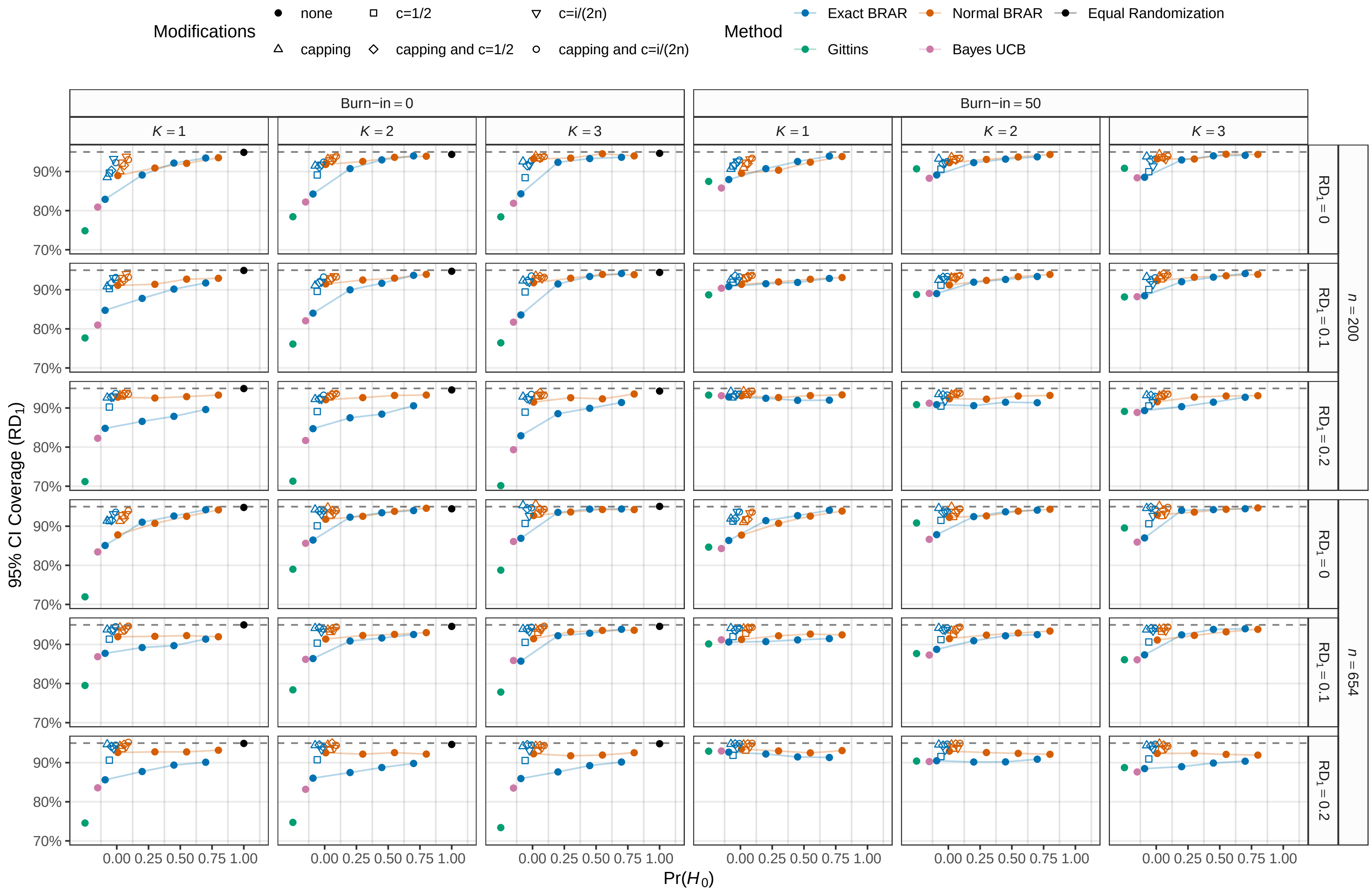

Figure S6: Empirical coverage of the 95% Wald confidence interval for the rate difference RD_1 based on 10′000 simulation repetitions. The maximum MCSE is 0.46%.

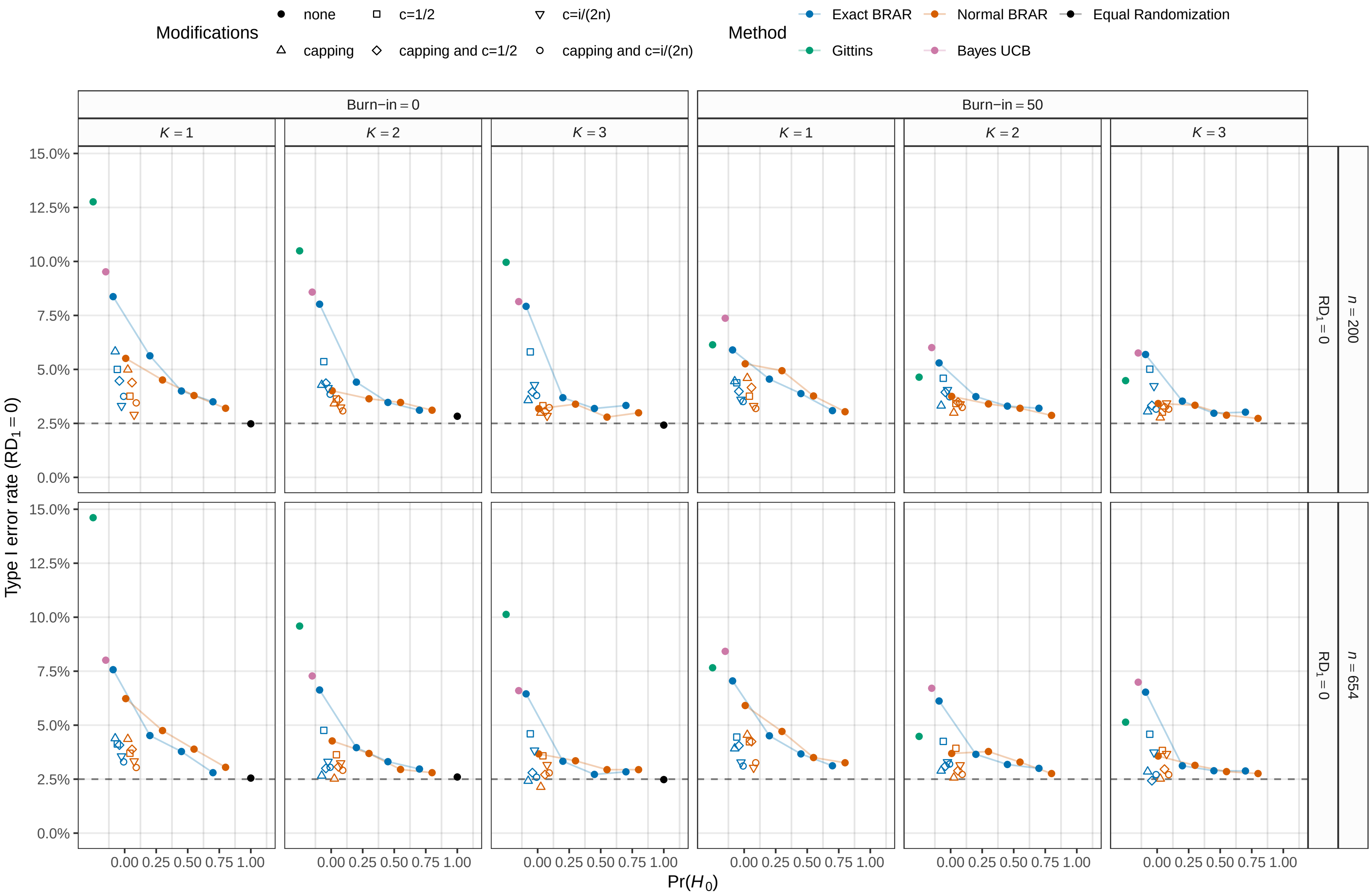

Figure S7: Empirical type I error rate of the Wald test of $\mathrm{RD}_1 = 0$ based on 10′000 simulation repetitions. The maximum MCSE is 0.35%.

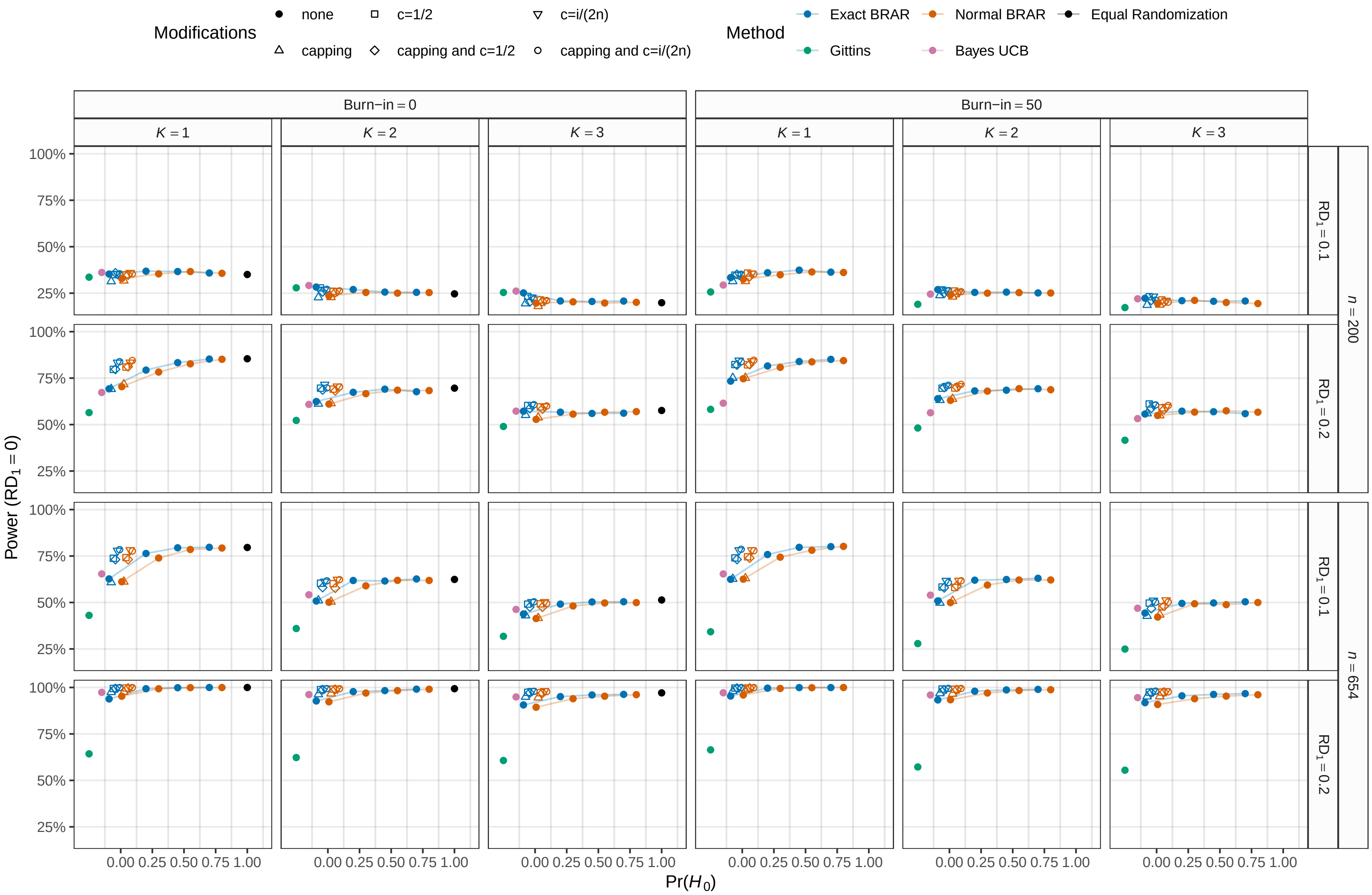

Figure S8: Empirical power of the Wald test of $\mathrm{RD}_1 = 0$ based on 10′000 simulation repetitions. The maximum MCSE is 0.5%.

modifications were applied. Similarly, increasing the prior probability of H_0 decreased negative imbalance, in some cases even below Thompson sampling with modifications. As in the simulation study from Robertson et al. (2023), negative imbalance of Thompson sampling and Bayesian RAR with $\Pr(H_0) < 1$ was higher than under equal randomization in conditions with small effect size $\mathrm{RD}_1 = 0.1$, but equal or lower than under equal randomization in conditions with large effect size $\mathrm{RD}_1 = 0.2$.

Bias and coverage Figures S5 and S6 show empirical bias and coverage related to estimates of the rate difference between the first treatment group and the control group RD_1. In conditions where there was no difference ($\mathrm{RD}_1 = 0$), all methods produced unbiased point estimates, though all methods but equal randomization also showed undercoverage. Bias occurred in conditions with non-zero rate differences, the bias being the greatest for the Gittins index, Bayes UCB, and Thompson sampling, and decreasing to some extent when modifications were introduced or the prior probability of H_0 increased. Similarly, modifications or increasing prior probabilitities improved coverage, although coverage still remained suboptimal in most conditions. For large rate difference conditions, coverage was often better for the normal than the exact version of RAR. Similarly, bias was in some cases somewhat larger for the exact compared to the normal version. This could potentially be due to the normal prior distribution on the log odds ratio being better tuned to effect sizes in the data-generating mechanism difference compared to the uniform priors on the probabilities under the binomial model.

Type I error rate and power Figures S7 and S8 show empirical type I error rate and power associated with the Wald test of $\mathrm{RD}_1 = 0$. We see that the Gittins index, Bayes UCB, and Thompson sampling methods showed an inflated type I error rate above the nominal 2.5% which was reduced to some extent by increasing either the prior probability of H_0 or applying modifications. In the same way, Gittins index, Bayes UCB, and Thompson sampling often exhibited reduced power compared to equal randomization, which was again alleviated by modifications or increasing $\Pr(H_0)$. In small sample sizes and large rate differences, power was slightly increased for the exact compared to the normal version of RAR, while for Thompson sampling the type I rate was slightly higher for the exact compared to the normal version.

Appendix D Software and data

Code and data to reproduce our analyses are openly available at `https://github.com/SamCH93/brar`. A snapshot of the repository at the time of writing is available at `https://doi.org/10.5281/zenodo.17248628`. We used the statistical programming language R version 4.5.2 (2025-10-31) for analyses (R Core Team, 2024) along with the `ggplot2` (Wickham, 2016), `dplyr` (Wickham et al., 2023), `SimDesign` (Chalmers and Adkins, 2020), `mvtnorm` (Genz and Bretz, 2009), `ggh4x` (van den Brand, 2024), `ggpubr` (Kassambara, 2023), `RARtrials` (Xu et al., 2025), and `knitr` (Xie, 2015) packages.

Computational details

```
cat(paste(Sys.time(), Sys.timezone(), "\n"))

## 2026-03-04 13:03:17.594902 Europe/Zurich

sessionInfo()

## R version 4.5.2 (2025-10-31)
## Platform: x86_64-pc-linux-gnu
## Running under: Ubuntu 24.04.4 LTS
##
## Matrix products: default
## BLAS:   /usr/lib/x86_64-linux-gnu/blas/libblas.so.3.12.0
## LAPACK: /usr/lib/x86_64-linux-gnu/lapack/liblapack.so.3.12.0  LAPACK version 3.12.0
##
## locale:
##  [1] LC_CTYPE=en_US.UTF-8       LC_NUMERIC=C
##  [3] LC_TIME=de_CH.UTF-8        LC_COLLATE=en_US.UTF-8
##  [5] LC_MONETARY=de_CH.UTF-8    LC_MESSAGES=en_US.UTF-8
##  [7] LC_PAPER=de_CH.UTF-8       LC_NAME=C
##  [9] LC_ADDRESS=C               LC_TELEPHONE=C
## [11] LC_MEASUREMENT=de_CH.UTF-8 LC_IDENTIFICATION=C
##
## time zone: Europe/Zurich
## tzcode source: system (glibc)
##
## attached base packages:
## [1] parallel  stats     graphics  grDevices utils     datasets  methods
## [8] base
##
## other attached packages:
## [1] Ternary_2.3.5  ggpubr_0.6.2   ggh4x_0.3.1    ggplot2_4.0.2  mvtnorm_1.3-3
## [6] brar_0.1       SimDesign_2.21 dplyr_1.2.0    knitr_1.50
##
## loaded via a namespace (and not attached):
##  [1] gtable_0.3.6        xfun_0.54           PlotTools_0.3.1
##  [4] rstatix_0.7.3       lattice_0.22-7      vctrs_0.7.1
##  [7] tools_4.5.2         generics_0.1.4      tibble_3.3.1
## [10] highr_0.11          pkgconfig_2.0.3     R.oo_1.27.1
## [13] RColorBrewer_1.1-3  S7_0.2.1            lifecycle_1.0.5
## [16] compiler_4.5.2      farver_2.1.2        brio_1.1.5
## [19] codetools_0.2-20    carData_3.0-5       httpuv_1.6.16
## [22] htmltools_0.5.9     Formula_1.2-5       later_1.4.4
## [25] pillar_1.11.1       car_3.1-3           tidyr_1.3.2
## [28] R.utils_2.13.0      sessioninfo_1.2.3   abind_1.4-8
## [31] mime_0.13           parallelly_1.46.1   tidyselect_1.2.1
```

```
## [34] digest_0.6.39       future_1.69.0       purrr_1.2.1
## [37] listenv_0.10.0      labeling_0.4.3      cowplot_1.2.0
## [40] fastmap_1.2.0       grid_4.5.2          cli_3.6.5
## [43] beepr_2.0           magrittr_2.0.4      broom_1.0.10
## [46] future.apply_1.20.1 clipr_0.8.0         withr_3.0.2
## [49] promises_1.5.0      scales_1.4.0        backports_1.5.0
## [52] sp_2.2-0            globals_0.18.0      audio_0.1-11
## [55] otel_0.2.0          gridExtra_2.3       progressr_0.18.0
## [58] ggsignif_0.6.4      R.methodsS3_1.8.2   shiny_1.12.1
## [61] pbapply_1.7-4       evaluate_1.0.5      RcppHungarian_0.3
## [64] testthat_3.3.0      viridisLite_0.4.3   rlang_1.1.7
## [67] Rcpp_1.1.1          xtable_1.8-4        glue_1.8.0
## [70] R6_2.6.1
```